CHAPTER 11

Spectroscopy

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: bd077e2122b86b13d968661aba92c085b7c260d29ba1367394b77a7914e2b687

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch11:exercise:ir-read | ir-read:stage-1, ir-read:stage-2, ir-read:stage-3

ch11:exercise:uv-energy | uv-energy:stage-1, uv-energy:stage-2, uv-energy:stage-3

ch11:exercise:nmr-ethyl | nmr-ethyl:stage-1, nmr-ethyl:stage-2, nmr-ethyl:stage-3

ch11:exercise:combined-evidence | combined-evidence:stage-1, combined-evidence:stage-2, combined-evidence:stage-3

ch11:lesson:ir | ir:rule, ir:cue, ir:trap, ir:app, ir:step-1, ir:step-2, ir:step-3

ch11:lesson:uv | uv:rule, uv:cue, uv:trap, uv:app, uv:step-1, uv:step-2

ch11:lesson:nmr | nmr:rule, nmr:cue, nmr:trap, nmr:app, nmr:step-1, nmr:step-2, nmr:step-3

ch11:lesson:combine | combine:rule, combine:cue, combine:trap, combine:app, combine:step-1, combine:step-2, combine:step-3

ch11:molecule:acetone | acetone

ch11:molecule:isopropanol | isopropanol

ch11:molecule:aceticacid | aceticacid

ch11:molecule:butadiene | butadiene

ch11:molecule:ethene | ethene

ch11:molecule:ethylacetate | ethylacetate

ch11:molecule:propanal | propanal

ch11:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 11.1 Infrared Spectroscopy

scope-2: 11.2 Ultraviolet Spectroscopy

scope-3: 11.3 Nuclear Magnetic Resonance Spectroscopy

Objective: Recognize approximate IR regions

Objective: Relate conjugation, photon energy, and absorbance

Objective: Interpret NMR shifts, integrals, and splitting

Supplied Kaplan curriculum: Chapter 11; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter11b.xhtml

IR functional-group regions: OpenStax 12.8 | https://openstax.org/books/organic-chemistry/pages/12-8-infrared-spectra-of-some-common-functional-groups

First-order proton splitting: OpenStax 13.6 | https://openstax.org/books/organic-chemistry/pages/13-6-spin-spin-splitting-in-1h-nmr-spectra

Conjugation and UV absorption: OpenStax 14.8 | https://openstax.org/books/organic-chemistry/pages/14-8-interpreting-ultraviolet-spectra-the-effect-of-conjugation

NMR-active nuclei: OpenStax 13.1, inspected | https://openstax.org/books/organic-chemistry/pages/13-1-nuclear-magnetic-resonance-spectroscopy

UV HOMO/LUMO excitation: OpenStax 14.7, inspected | https://openstax.org/books/organic-chemistry/pages/14-7-ultraviolet-spectroscopy

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

ir | IR: recognize functional groups

Vibrations must change dipole moment

ir:rule | Core Rules

IR probes molecular vibrations. A vibration absorbs IR when it changes the molecular dipole moment. Useful approximate regions: alcohol O–H 3200–3600 cm^{-1} (often broad); acid O–H 2500–3300 (very broad); C=O about 1650–1800; nitrile about 2210–2260. N–H is often near 3300, generally less broad than hydrogen-bonded OH.

ir:cue | Memory Cues

Use a band pattern, not one magic number. The fingerprint region below about 1500 cm^{-1} helps comparison.

ir:trap | Common Mistakes / MCAT Traps

Carbonyl positions vary with functional group, conjugation, ring strain, and environment. Absence of an OH band may support a structure, but sample conditions matter. IR alone seldom gives complete connectivity.

ir:app | MCAT Applications

Follow loss of a carbonyl band and appearance of OH when monitoring a carbonyl reduction, together with other evidence.

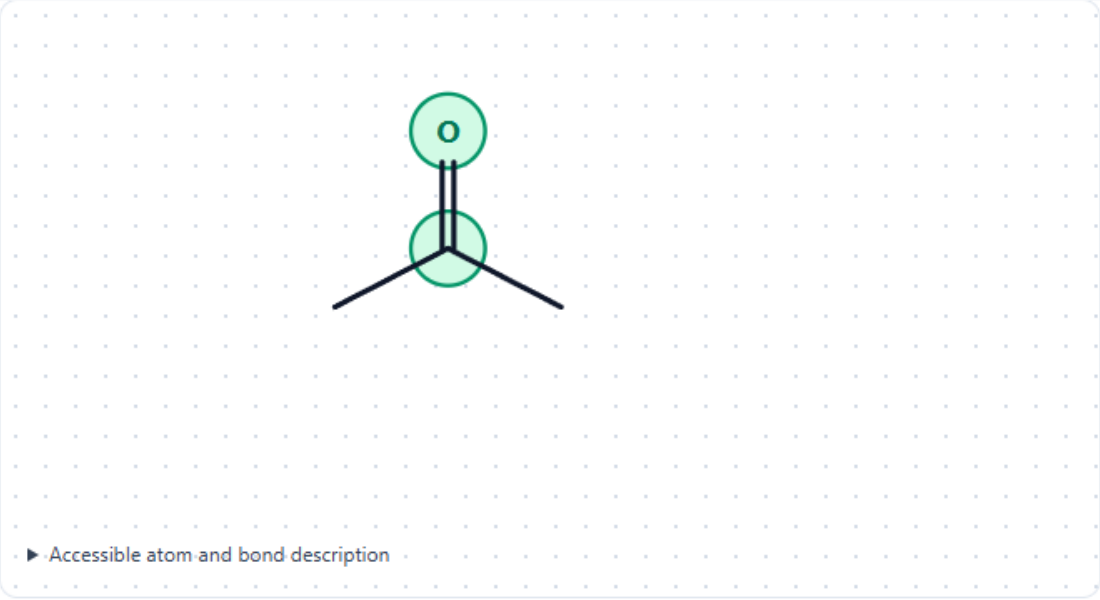
ir:step-1 | 1. Carbonyl signature

A simple saturated acyclic ketone often absorbs near 1715 cm^{-1}. The illustrative plot marks a band position, not an experimentally measured spectrum.

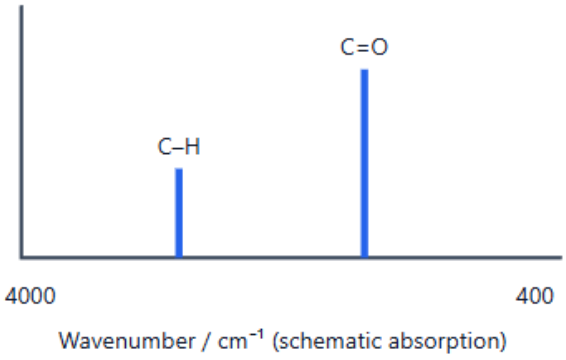
Displayed molecule ID: acetone

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Screenshot: ir-step-1



Screenshot: ir-step-1-diagram



Schematic ketone absorption; not measured data. Real IR spectra often plot transmittance downward.

Illustrative spectrum, not experimental data. Band widths, line spacing and heights are schematic; signal integration is not peak height.

{"diagramSpecification": {"type": "spectrum", "min": 400, "max": 4000, "unit": "Wavenumber / cm^{-1} (schematic absorption)", "caption": "Schematic ketone absorption; not measured data. Real IR spectra often plot transmittance downward.", "peaks": [{"x": 1715, "height": 0.95, "label": "C=O"}, {"x": 2950, "height": 0.45, "label": "C–H"}]}}

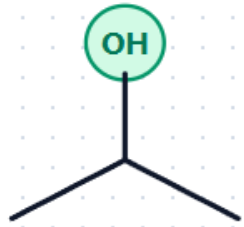
ir:step-2 | 2. Compare OH

An alcohol has an O–H vibration often broadened by hydrogen bonding. Converting acetone to propan-2-ol changes both the C=O and O–H evidence.

Displayed molecule ID: isopropanol

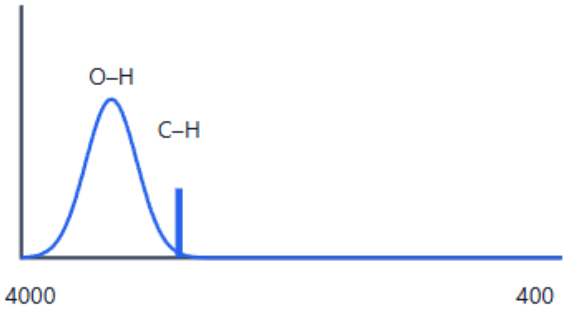
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Screenshot: ir-step-2



▶ Accessible atom and bond description

Screenshot: ir-step-2-diagram



Schematic hydrogen-bonded alcohol: broad OH and no ketone carbonyl. Widths and intensities are illustrative.

Illustrative spectrum, not experimental data. Band widths, line spacing and heights are schematic; signal integration is not peak height.

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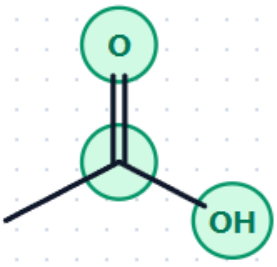
ir:step-3 | 3. Acid pattern

A carboxylic acid can show both a carbonyl band and a very broad OH band extending across roughly 2500–3300 cm^{-1}. Interpret them together.

Displayed molecule ID: aceticacid

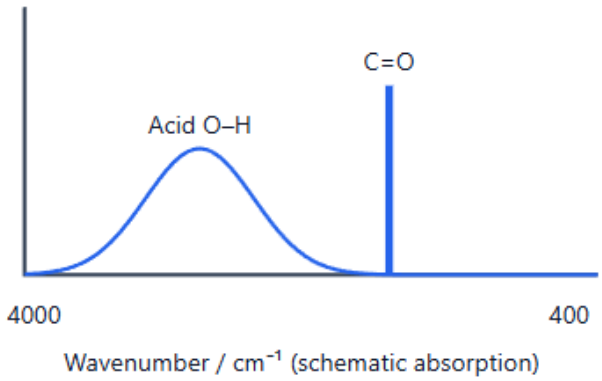
{"highlights": {"prin": [1, 2, 3]}, "revealName": true}

Screenshot: ir-step-3



▶ Accessible atom and bond description

Screenshot: ir-step-3-diagram



Schematic carboxylic acid: very broad OH together with C=O; not measured data.

Illustrative spectrum, not experimental data. Band widths, line spacing and heights are schematic; signal integration is not peak height.

{"diagramSpecification": {"type": "spectrum", "min": 400, "max": 4000, "unit": "Wavenumber / cm^{-1} (schematic absorption)", "caption": "Schematic carboxylic acid: very broad OH together with C=O; not measured data.", "peaks": [{"x": 2900, "height": 0.6, "label": "Acid O–H", "shape": "broad", "width": 800}, {"x": 1710, "height": 0.9, "label": "C=O"}]}}

uv | UV–visible: conjugation and energy

Longer wavelength means lower photon energy

uv:rule | Core Rules

UV–visible absorption promotes electrons between electronic states. In many comparable conjugated pi systems, increasing conjugation lowers the transition energy and shifts absorption to longer wavelength. Photon energy obeys E=hc/λ. Beer–Lambert law is A=εlc in its linear regime. For a simple pi system, a pi → pi* excitation can promote an electron from the highest occupied molecular orbital (HOMO) to the lowest unoccupied molecular orbital (LUMO). Extended conjugation often narrows that gap in comparable systems; allowed transitions and actual spectra also depend on the molecule.

uv:cue | Memory Cues

More extended conjugation often gives a red shift. Doubling wavelength halves photon energy.

uv:trap | Common Mistakes / MCAT Traps

Conjugation trends compare appropriate related systems, not every possible molecule. UV absorption does not uniquely identify a structure. Beer–Lambert requires compatible units and an appropriate concentration range.

uv:app | MCAT Applications

Conjugated chromophores underlie retinal photochemistry and many biological absorbance measurements.

uv:step-1 | 1. Follow conjugation

Alternating C=C–C=C bonds create a connected p-orbital system. Adjacent p orbitals must remain sufficiently aligned for conjugation.

Displayed molecule ID: butadiene

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Screenshot: uv-step-1



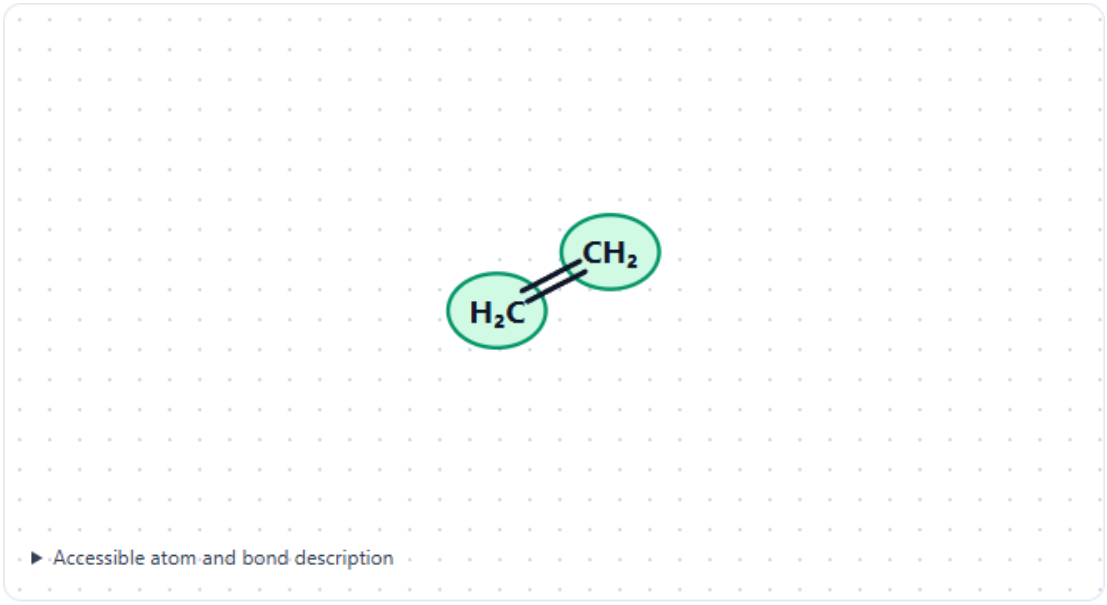
uv:step-2 | 2. Connect wavelength to energy

Relative to an isolated comparable alkene, conjugation often reduces the pi-to-pi-star transition energy. Use E=hc/λ to translate that smaller gap into longer wavelength.

Displayed molecule ID: ethene

{"highlights": {"prin": [0, 1]}, "revealName": true}

Screenshot: uv-step-2



nmr | Proton NMR: shift, integration, splitting

Three independent kinds of evidence

nmr:rule | Core Rules

Chemical shift reports environment; integration reports relative proton count; splitting reports coupling. In simple first-order cases, coupling to n equivalent neighboring protons produces n+1 lines. Equivalent protons do not split one another. Deshielded protons appear downfield at higher ppm.

nmr:cue | Memory Cues

Shift = neighborhood; area = population; splitting = neighbors. Upfield means lower ppm and greater shielding; downfield means higher ppm. Tetramethylsilane (TMS) is commonly assigned $\delta = 0$ ppm. Coupling constant J is the line spacing in Hz, not the integral or chemical shift.

nmr:trap | Common Mistakes / MCAT Traps

n+1 is a limited first-order rule, not universal. Nonequivalent neighbors can give more complex patterns; exchangeable OH/NH protons often do not show the expected simple coupling. Chemical equivalence depends on symmetry and timescale. NMR requires nonzero nuclear spin. Odd-mass isotopes such as ^{1}H and ^{13}C are active; odd-atomic-number nuclei with even mass, such as ^{2}H and ^{14}N, can also be active. Even-proton/even-neutron nuclei such as ^{12}C and ^{16}O have spin zero. Some active isotopes have broad or difficult signals, so activity alone does not guarantee easy detection.

nmr:app | MCAT Applications

Typical rough proton regions: alkyl 0–3 ppm; C next to O often 3–4.5; vinylic about 4.5–6.5; aromatic 6–8.5; aldehydic 9–10; acid OH often 10–13. Use the full pattern.

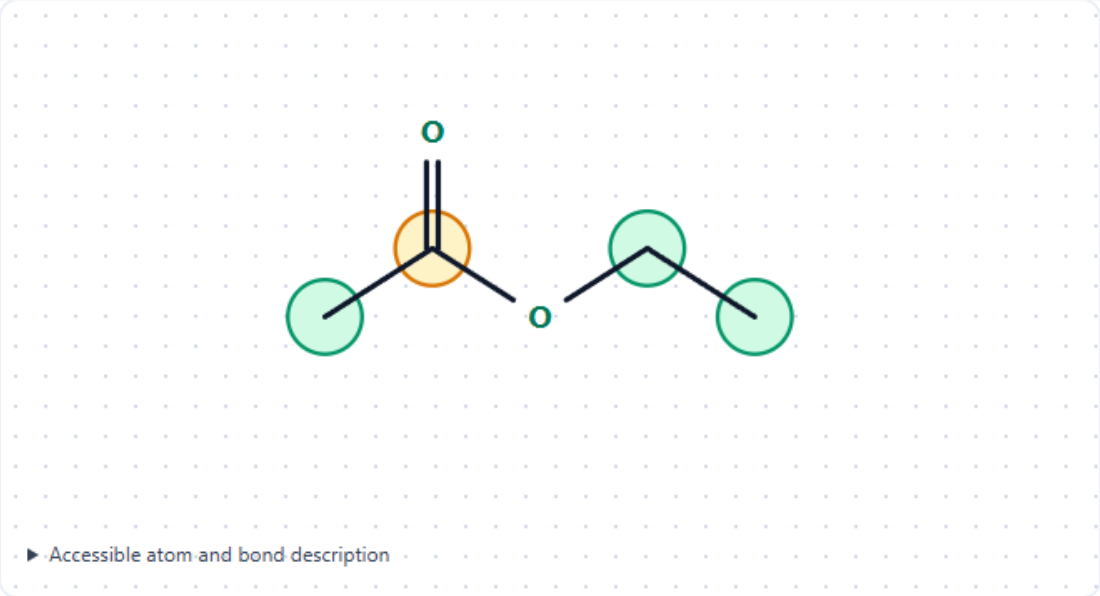
nmr:step-1 | 1. Three proton sets

Ethyl ethanoate has acyl methyl, OCH2, and terminal ethyl methyl proton sets. Their integrals are 3:2:3. The carbonyl carbon has no H. The three green carbon sites carry separate proton environments, not one combined set. Their positions and integrals distinguish them; the orange carbonyl carbon carries no H.

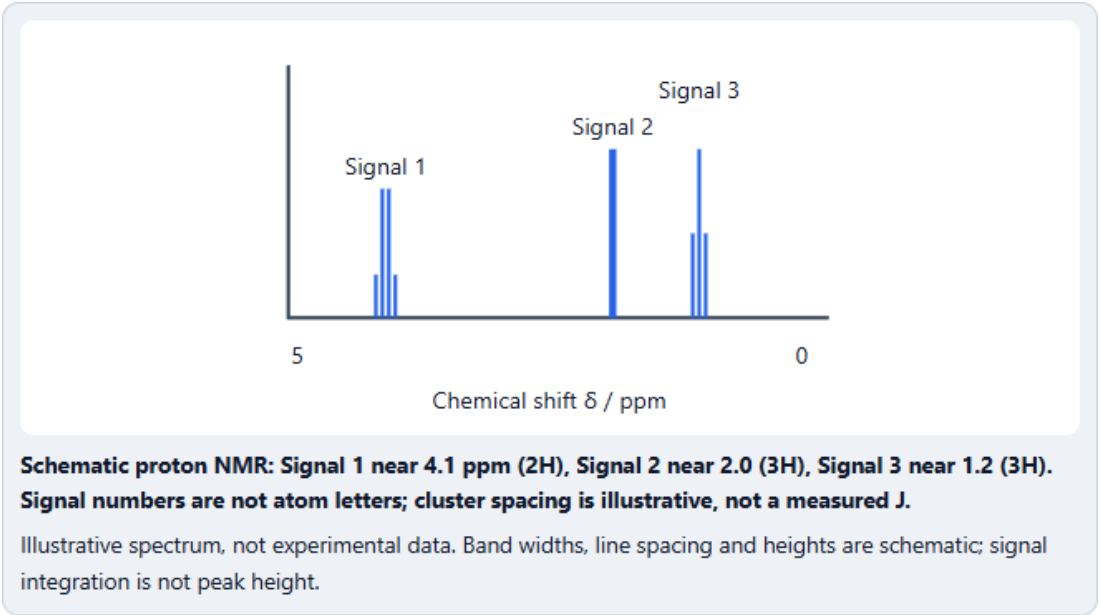
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Screenshot: nmr-step-1



Screenshot: nmr-step-1-diagram



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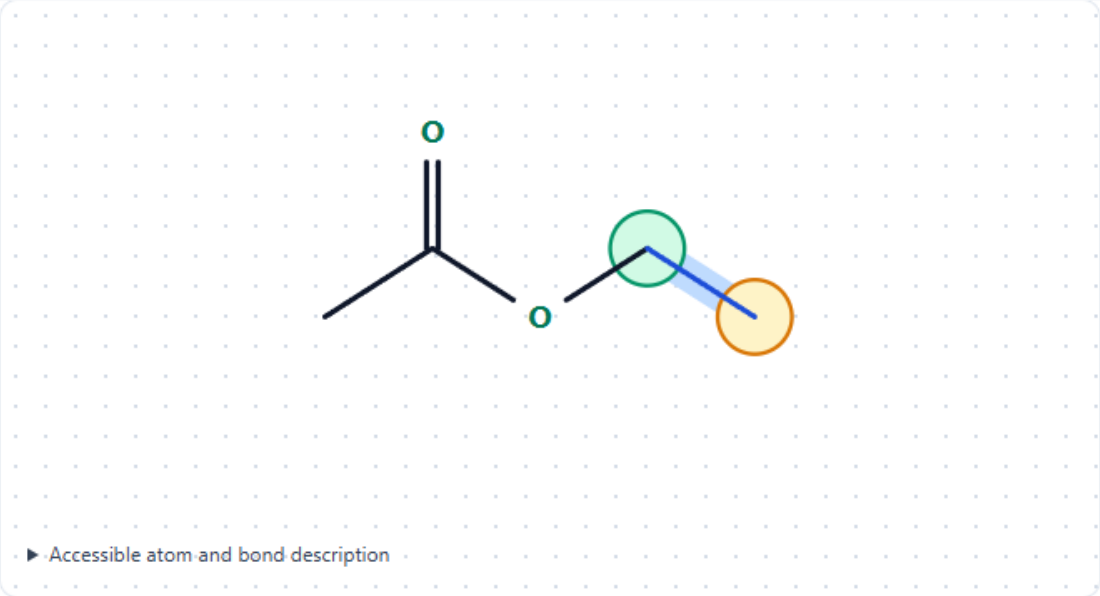
nmr:step-2 | 2. Ethyl splitting

OCH2 couples to three neighboring methyl H atoms, giving a quartet in a simple first-order picture. The terminal CH3 couples to two methylene H atoms, giving a triplet.

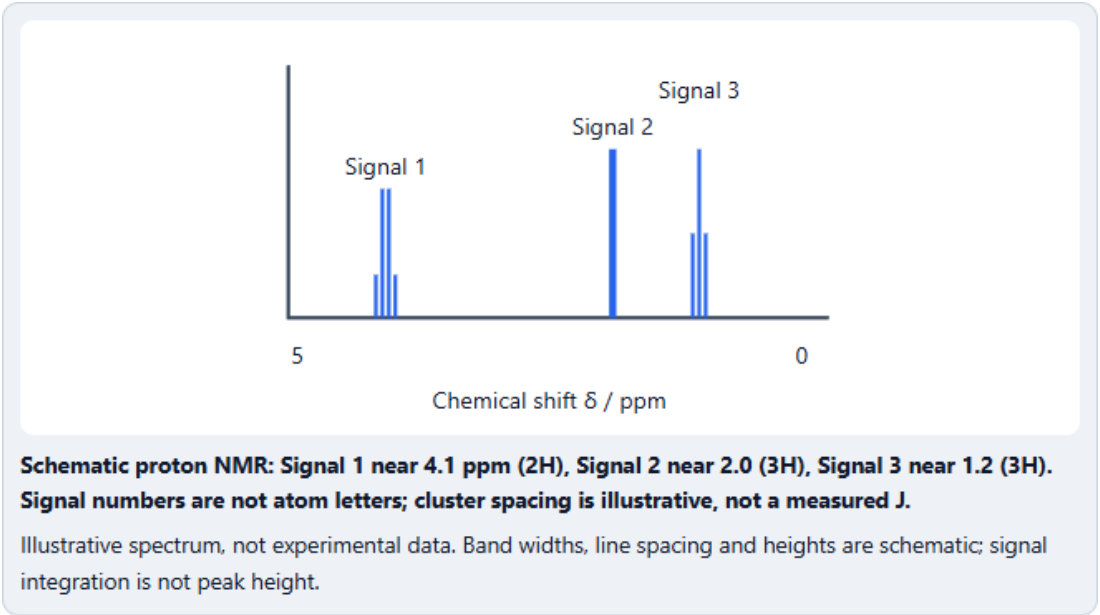
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Screenshot: nmr-step-2



Screenshot: nmr-step-2-diagram



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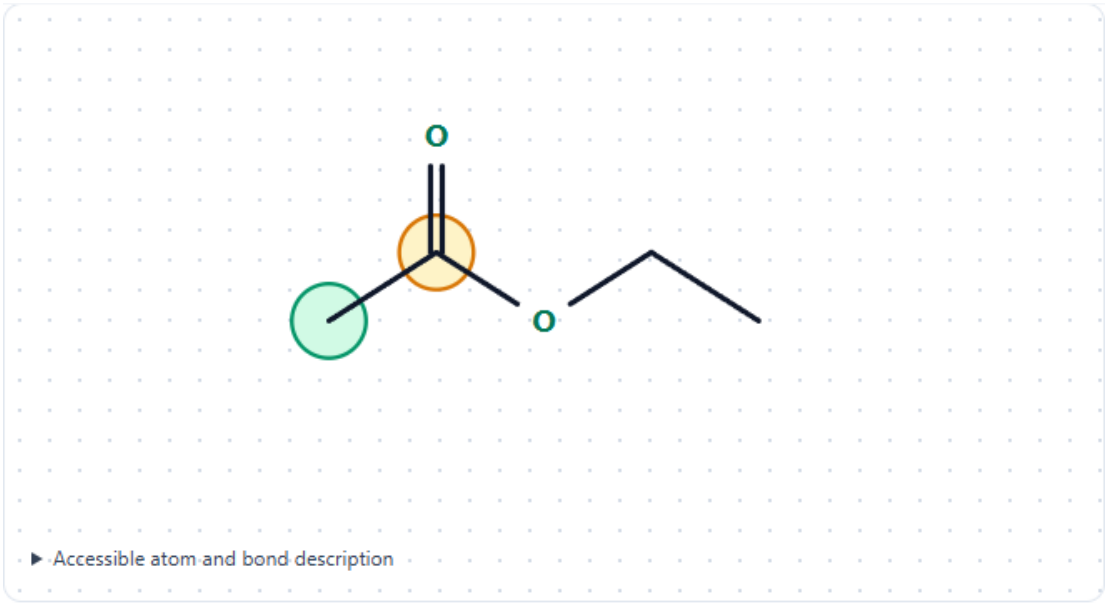
nmr:step-3 | 3. Acyl methyl

The acyl methyl has no H-bearing carbon immediately across the carbonyl bond, so it is approximately a singlet in this example. Do not apply n+1 by counting every H anywhere in the molecule.

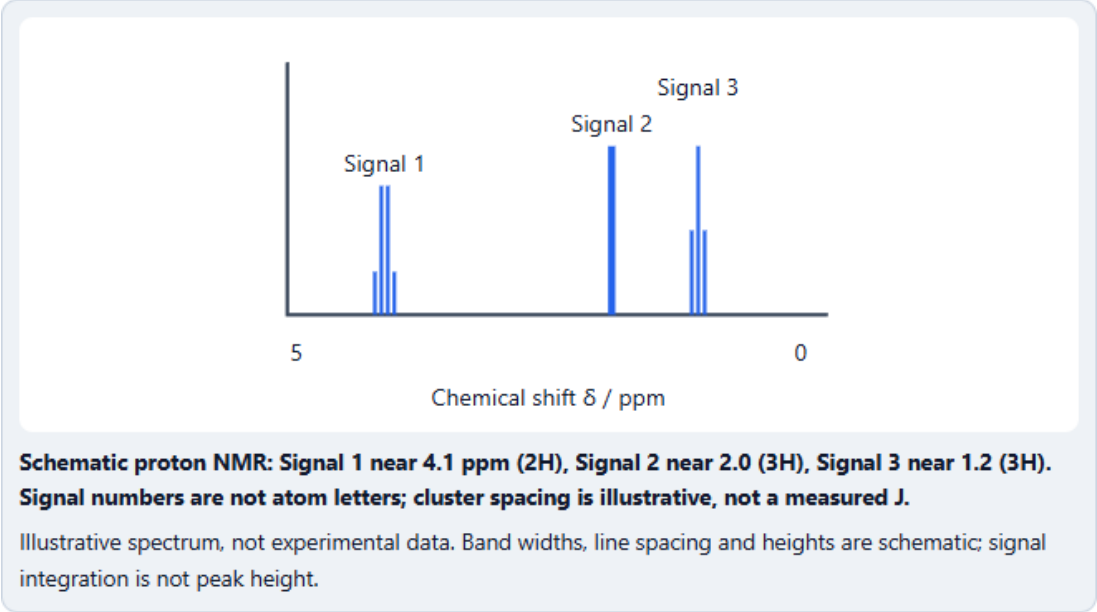
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Screenshot: nmr-step-3



Screenshot: nmr-step-3-diagram



{"diagramSpecification": {"type": "spectrum", "min": 0, "max": 5, "unit": "Chemical shift δ / ppm", "caption": "Schematic proton NMR: Signal 1 near 4.1 ppm (2H), Signal 2 near 2.0 (3H), Signal 3 near 1.2 (3H). Signal numbers are not atom letters; cluster spacing is illustrative, not a measured J.", "peaks": [{"x": 4.1, "height": 0.65, "label": "Signal 1", "lines": 4, "spacing": 0.06}, {"x": 2, "height": 0.85, "label": "Signal 2", "lines": 1}, {"x": 1.2, "height": 0.85, "label": "Signal 3", "lines": 3, "spacing": 0.06}]}}

combine | Combine formula, IR, and NMR

Independent clues should agree

combine:rule | Core Rules

For ordinary neutral closed-shell organic formulas, degree of unsaturation (DBE) = (2C+2+N−H−X)/2; O and divalent S do not enter. A ring or double bond contributes one DBE; a triple bond contributes two. Then combine IR group evidence with NMR environments and integrations.

combine:cue | Memory Cues

Formula constrains possibilities; IR finds groups; NMR organizes the skeleton.

combine:trap | Common Mistakes / MCAT Traps

A single NMR signal can represent several equivalent groups. DBE says how much unsaturation exists, not where it is. Do not force phosphorus-rich or unusual-valence formulas into this simple equation.

combine:app | MCAT Applications

Passage-based structure identification rewards eliminating structures inconsistent with even one reliable observation.

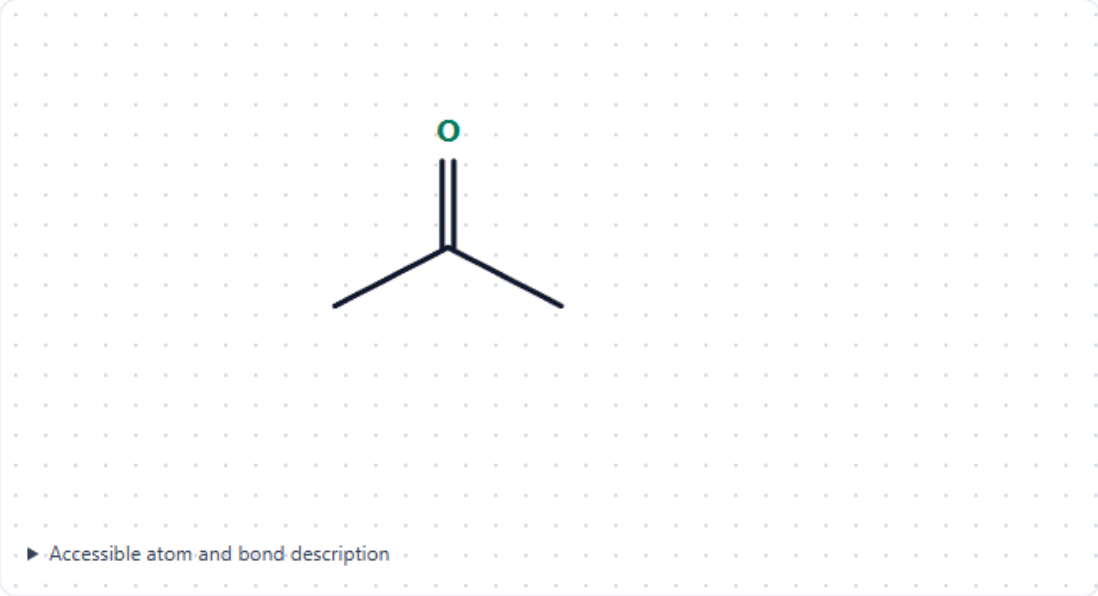
combine:step-1 | 1. Formula constraint

C3H6O has DBE=(2×3+2−6)/2=1. A carbonyl can account for that entire unit of unsaturation.

Displayed molecule ID: acetone

{"highlights": {}, "revealName": false}

Screenshot: combine-step-1



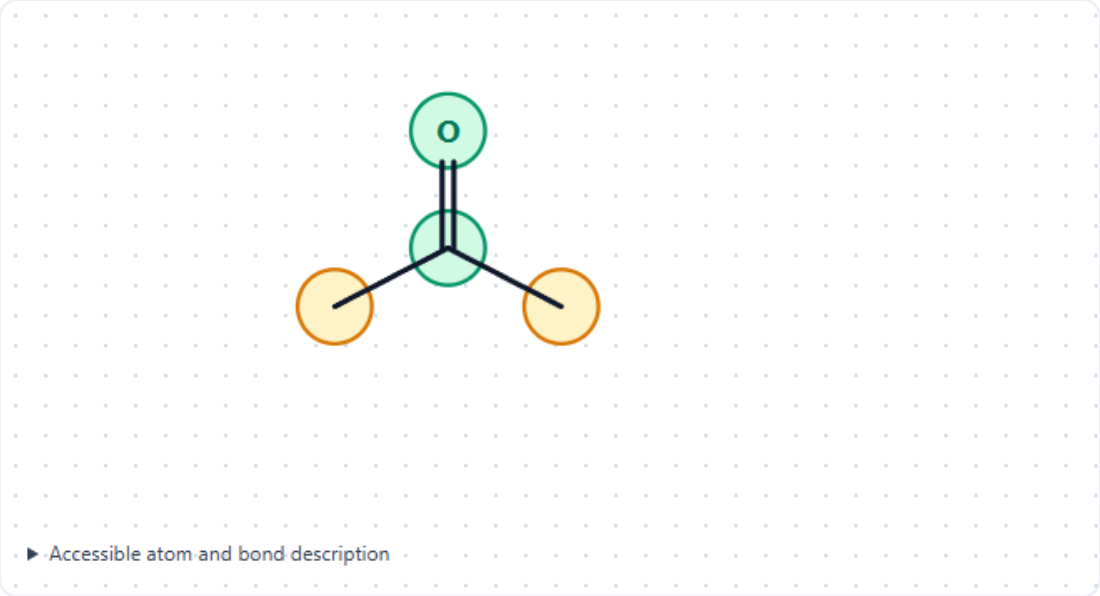
combine:step-2 | 2. Add two measurements

Suppose IR shows a ketone-like C=O band and proton NMR shows a single 6H singlet near 2 ppm. Two symmetry-equivalent methyl groups adjacent to C=O fit those data.

Displayed molecule ID: acetone

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Screenshot: combine-step-2



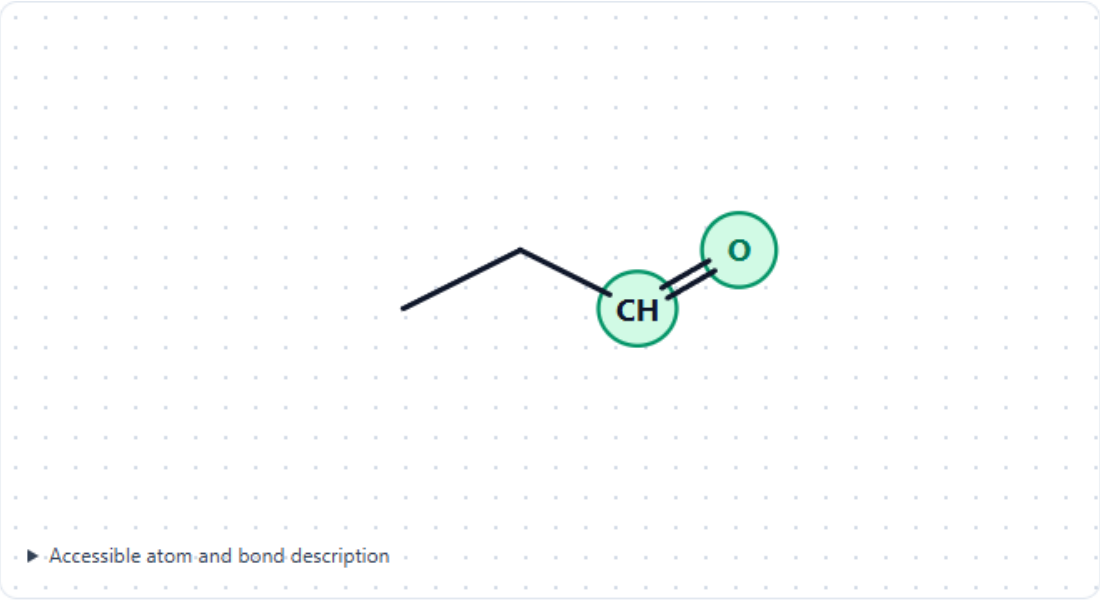
combine:step-3 | 3. Challenge an alternative

Propanal shares C3H6O and DBE=1 with acetone, and both have C=O absorption. But propanal has an aldehydic H (shown explicitly as CH at the terminal carbonyl carbon) and multiple proton environments; it does not fit the single 6H singlet. Compare every observation, not just formula and IR.

Displayed molecule ID: propanal

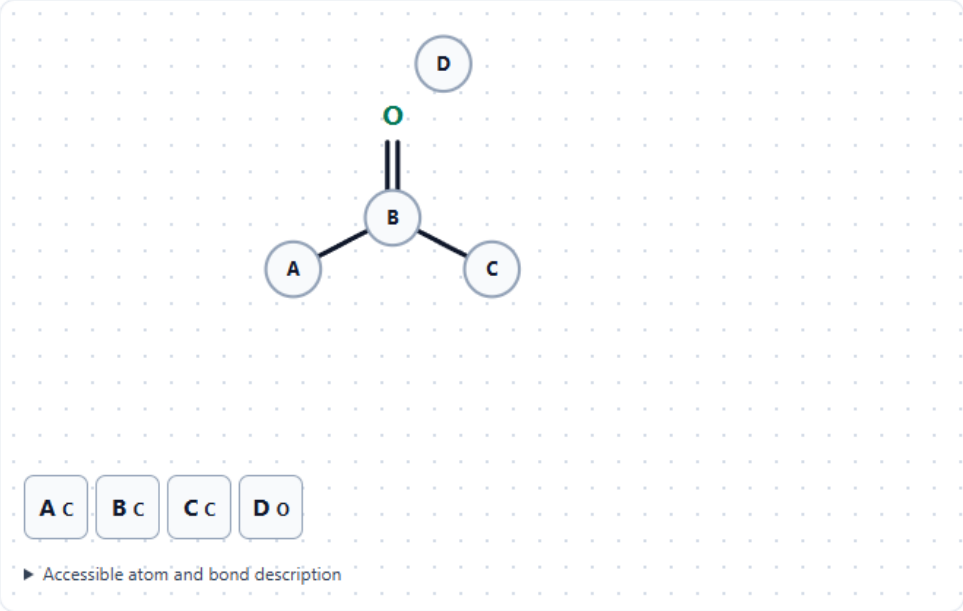
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Screenshot: combine-step-3

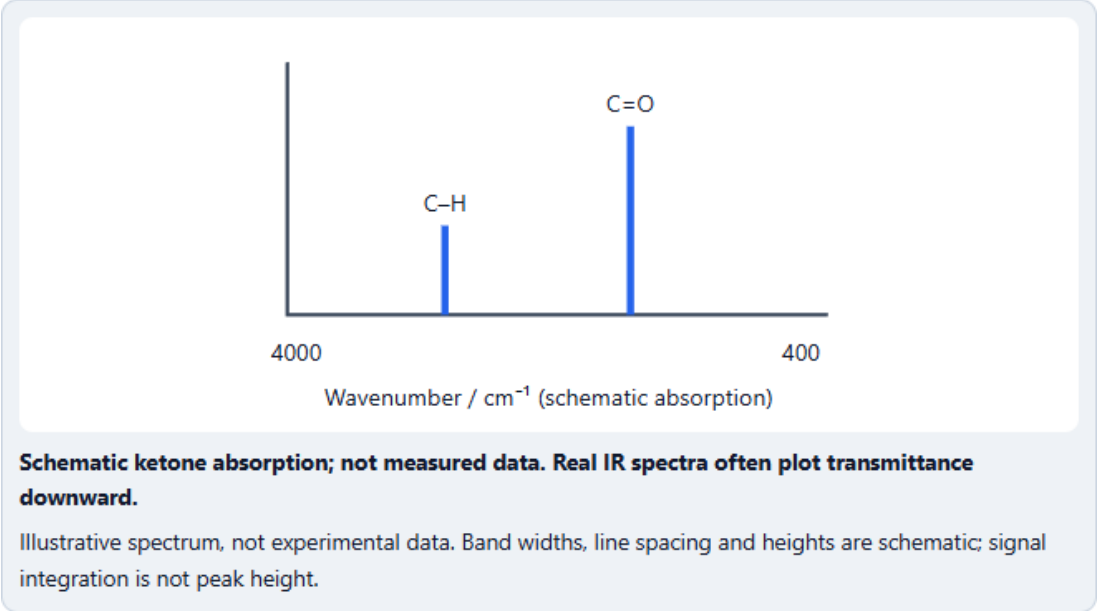


ir-read | Interpret functional-group evidence

Screenshot: ir-read-stage-1-molecule



Screenshot: ir-read-diagram

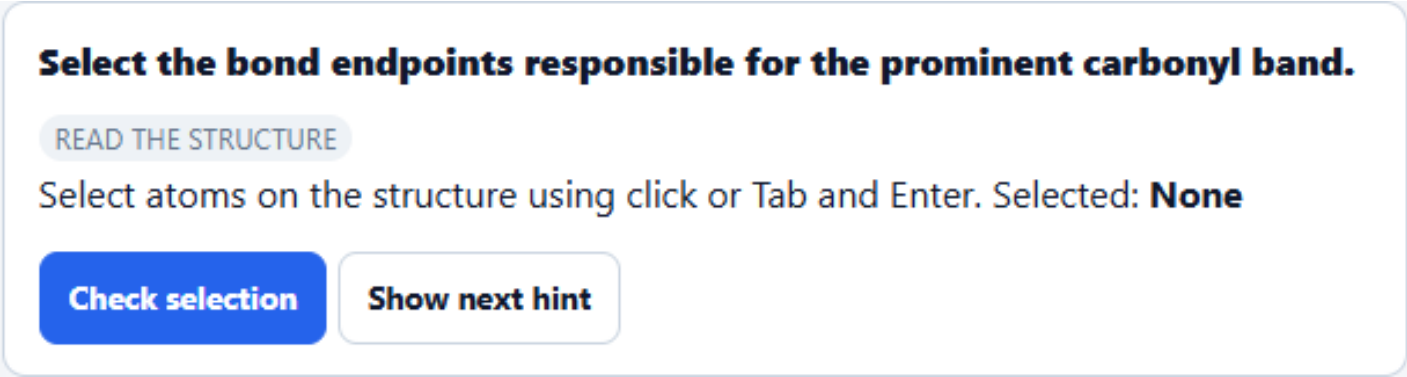


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ir-read:stage-1 | Select the bond endpoints responsible for the prominent carbonyl band.

Answer type: atoms

Screenshot: ir-read-stage-1-question



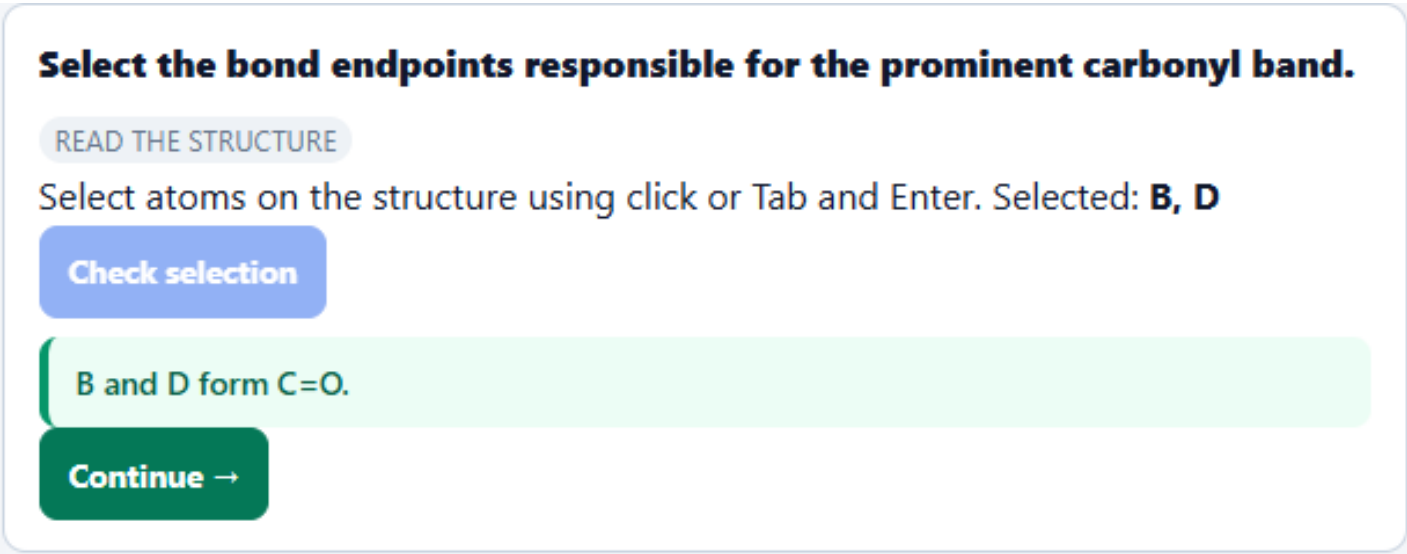
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Hint 1: Find the double bond to O.

Hint 2: Select both carbon and oxygen.

Explanation: B and D form C=O.

Screenshot: ir-read-stage-1-feedback



ir-read:stage-2 | An IR-active vibration must change...

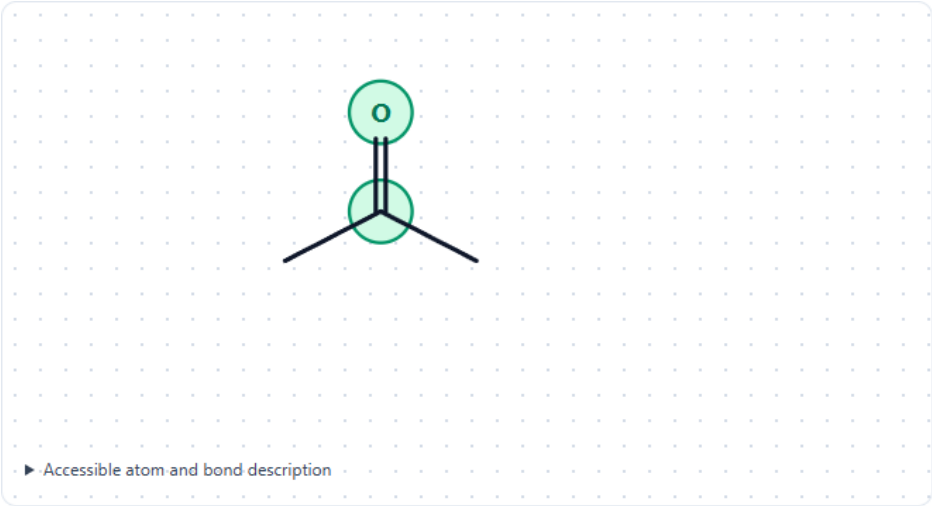
Answer type: choice

Option 0 (A): Dipole moment

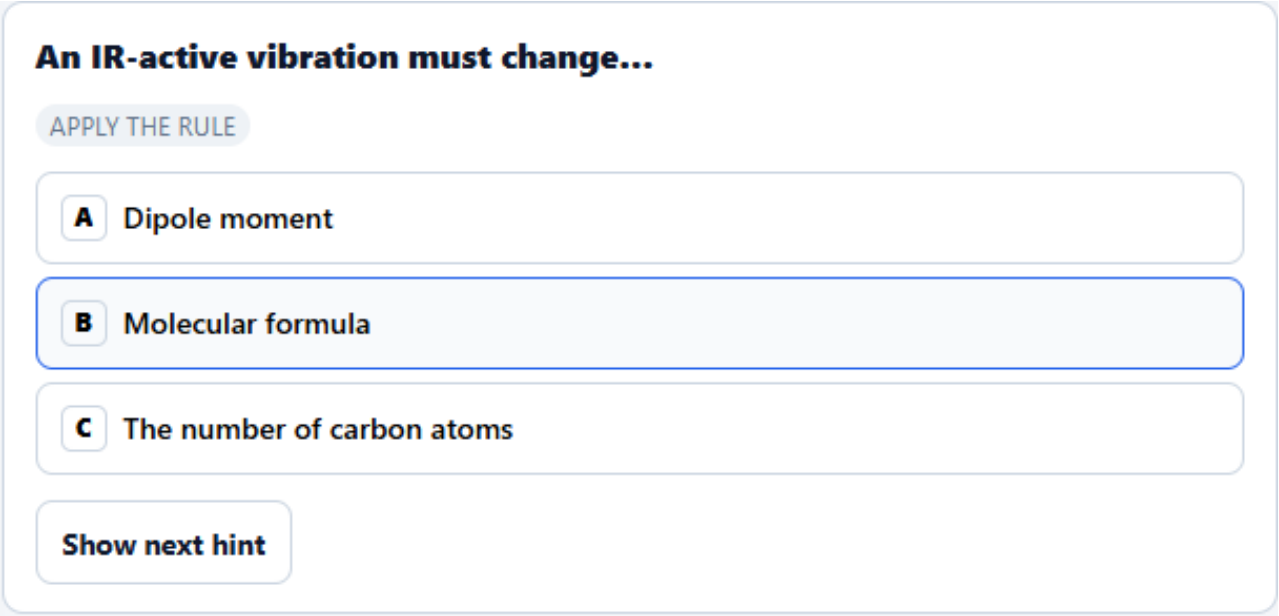
Option 1 (B): Molecular formula

Option 2 (C): The number of carbon atoms

Screenshot: ir-read-stage-2-molecule



Screenshot: ir-read-stage-2-question



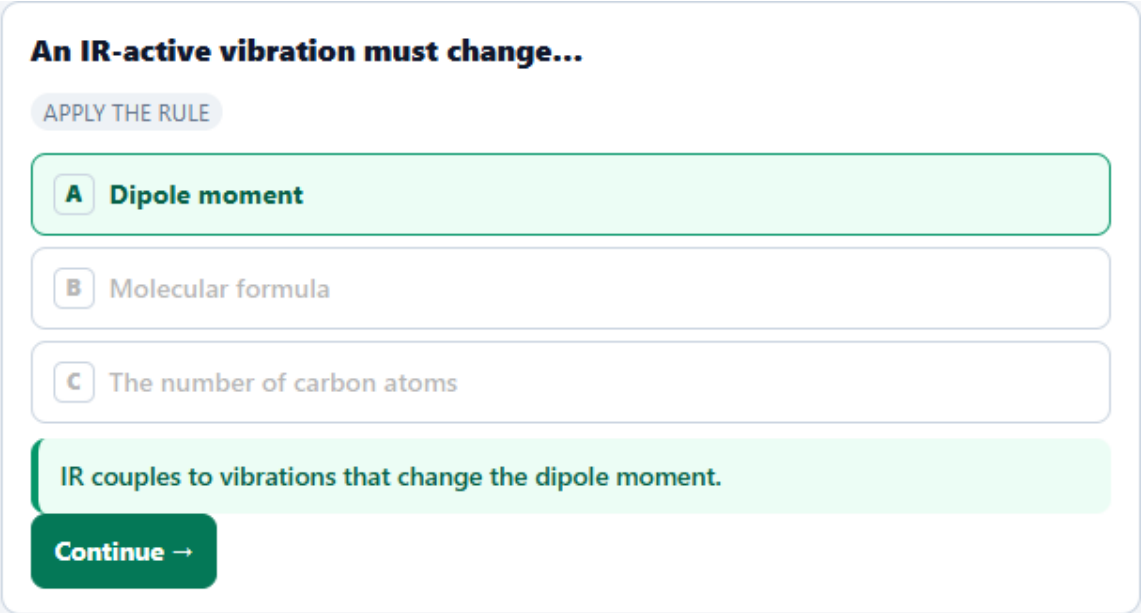
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Think about interaction with oscillating electric fields.

Hint 2: Not every possible vibration absorbs IR.

Explanation: IR couples to vibrations that change the dipole moment.

Screenshot: ir-read-stage-2-feedback



This stage uses a different molecule (connectivity shown).

ir-read:stage-3 | A broad 3200–3600 cm^{-1} band appearing after reduction supports...

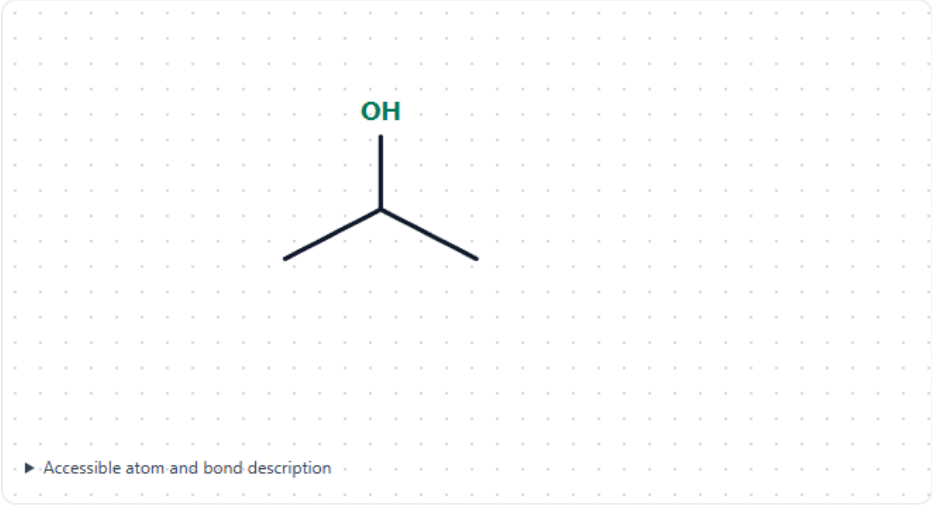
Answer type: choice

Option 0 (A): An alcohol OH

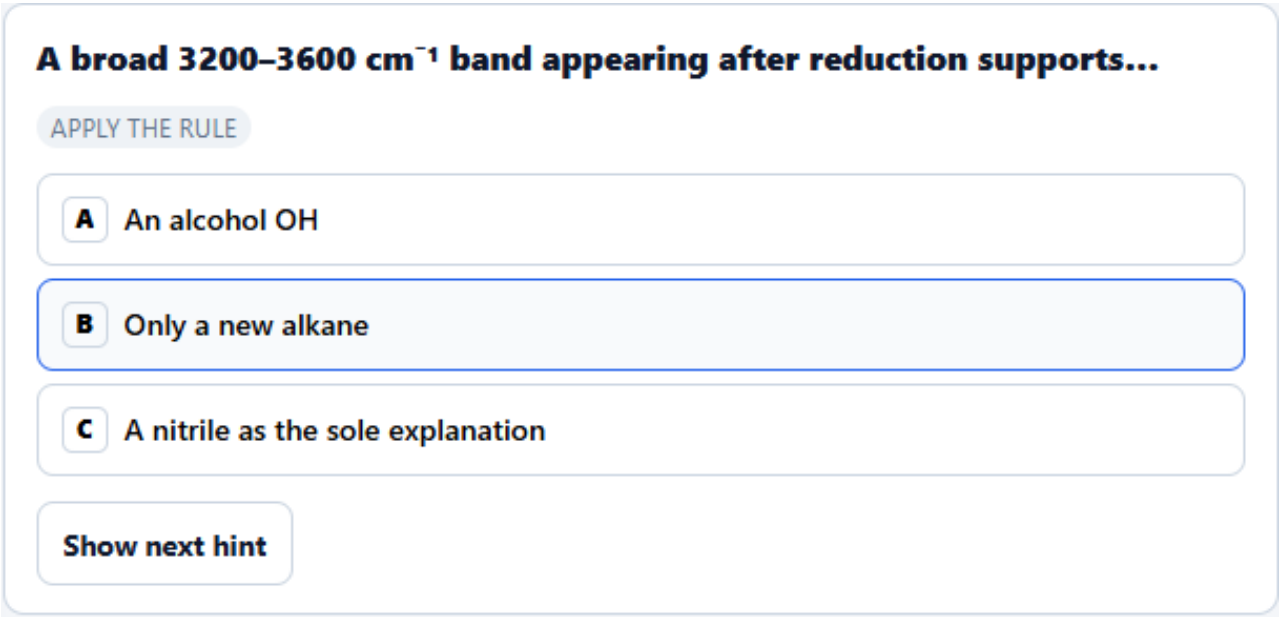
Option 1 (B): Only a new alkane

Option 2 (C): A nitrile as the sole explanation

Screenshot: ir-read-stage-3-molecule



Screenshot: ir-read-stage-3-question



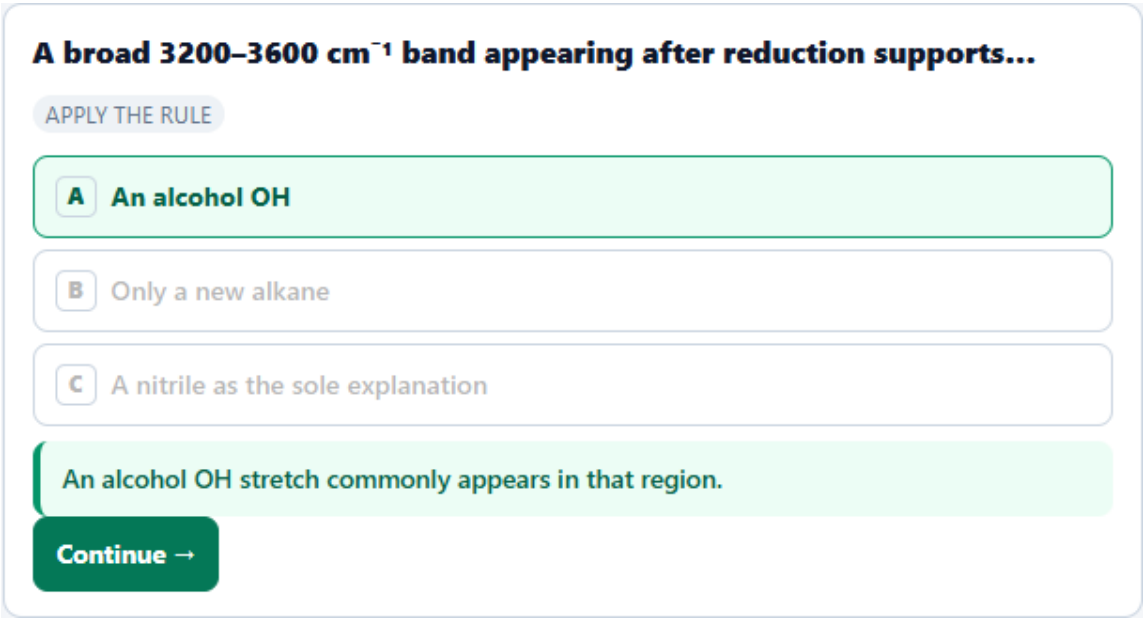
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Reduction retains oxygen.

Hint 2: C=O becomes C–OH.

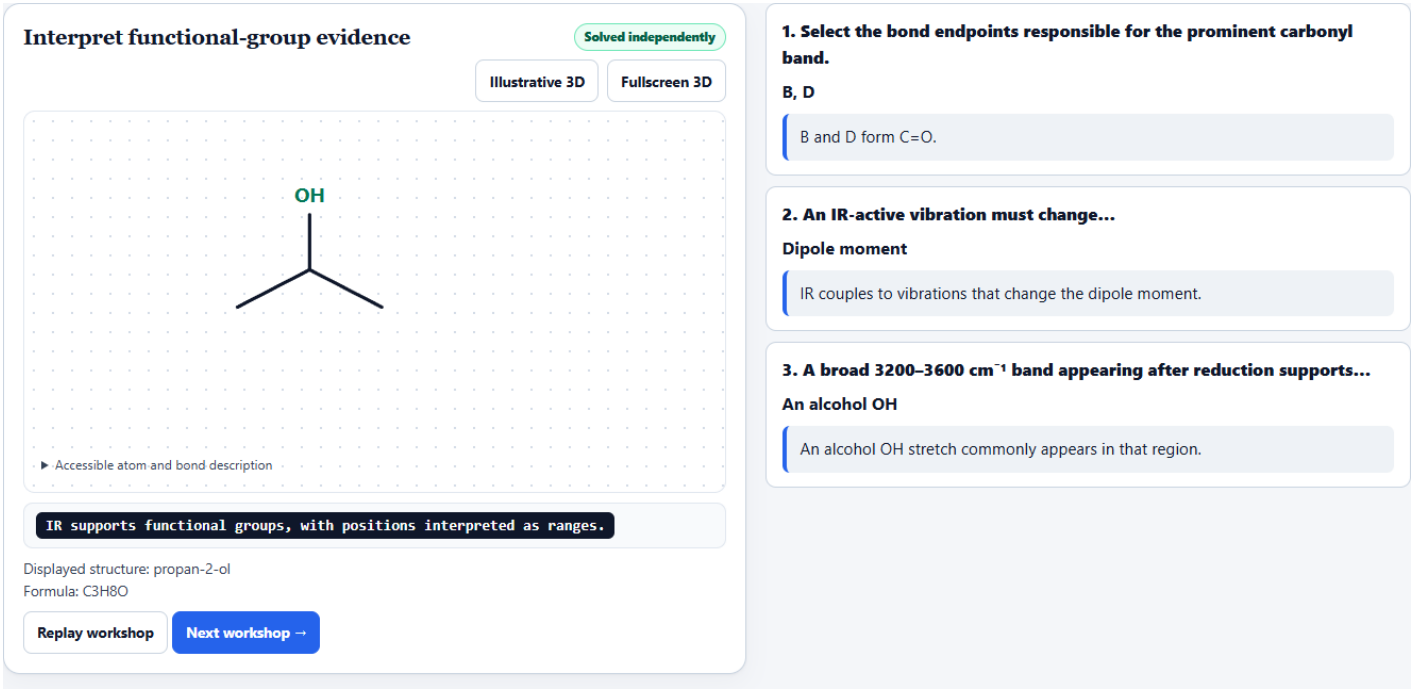
Explanation: An alcohol OH stretch commonly appears in that region.

Screenshot: ir-read-stage-3-feedback



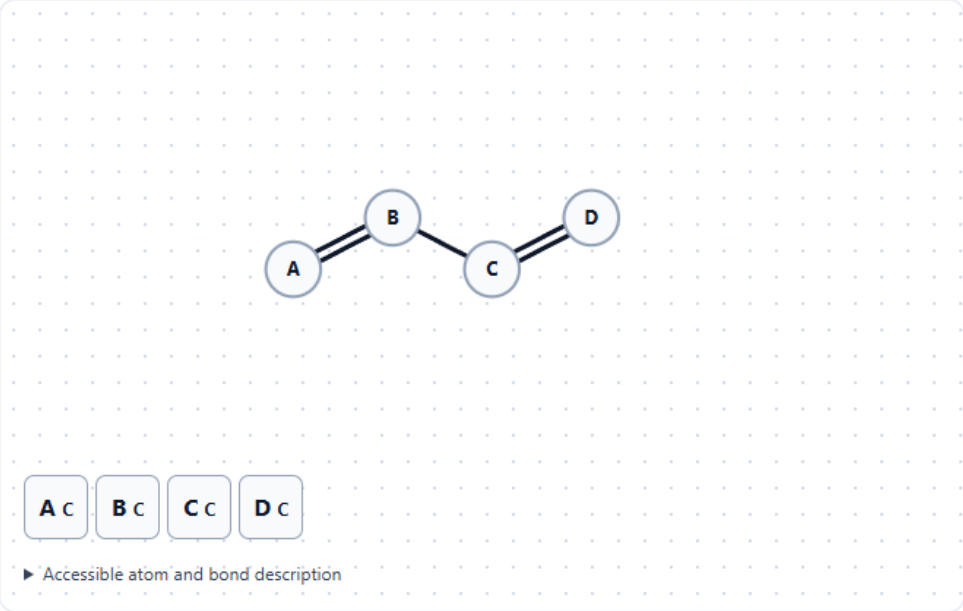
Completion summary: IR supports functional groups, with positions interpreted as ranges.

Screenshot: ir-read-complete



uv-energy | Connect conjugation and absorbance

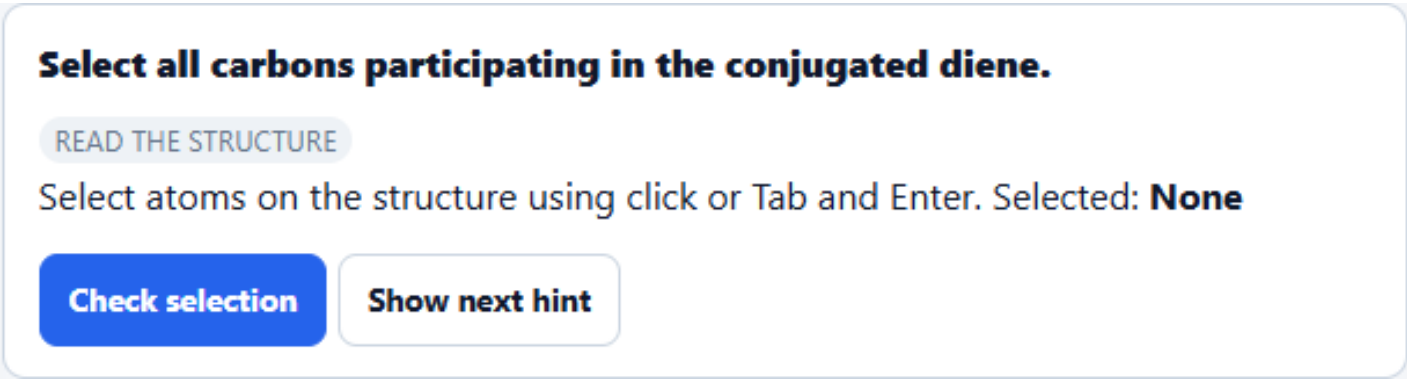
Screenshot: uv-energy-stage-1-molecule



uv-energy:stage-1 | Select all carbons participating in the conjugated diene.

Answer type: atoms

Screenshot: uv-energy-stage-1-question



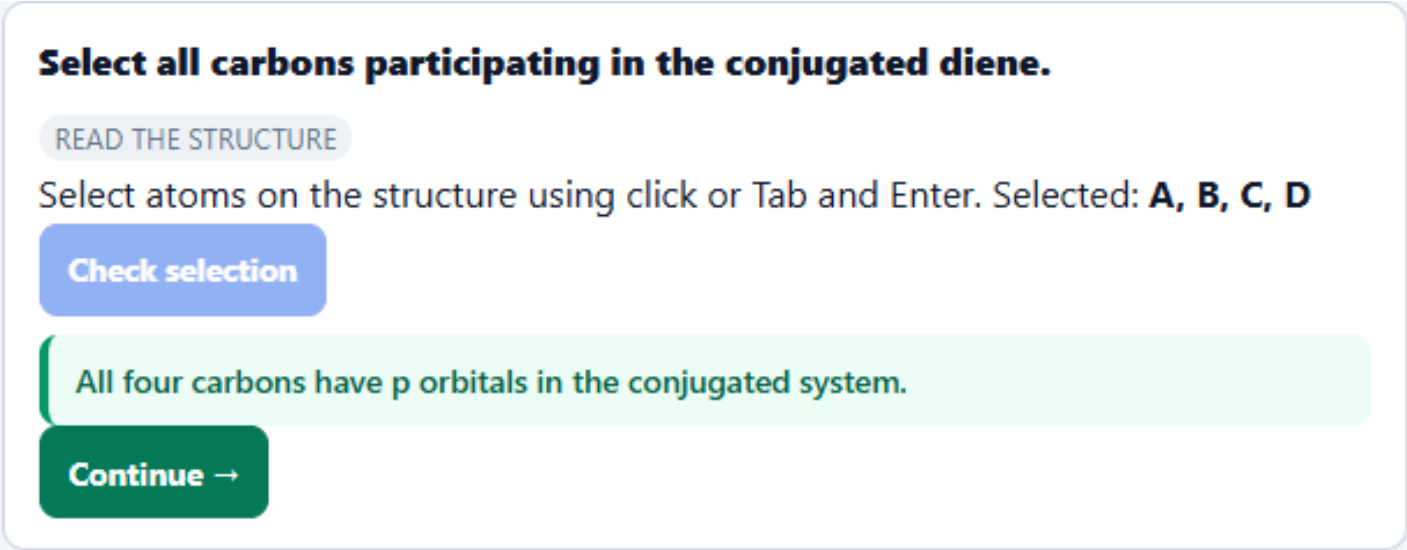
{"authorAnswer": [0, 1, 2, 3], "tolerance": null, "highlightsAfterCorrectAnswer": {"path": [0, 1, 2, 3]}}

Hint 1: Follow the alternating double–single–double pattern.

Hint 2: Include the carbons of both double bonds.

Explanation: All four carbons have p orbitals in the conjugated system.

Screenshot: uv-energy-stage-1-feedback



uv-energy:stage-2 | For comparable pi systems, increased conjugation often shifts absorption to...

Answer type: choice

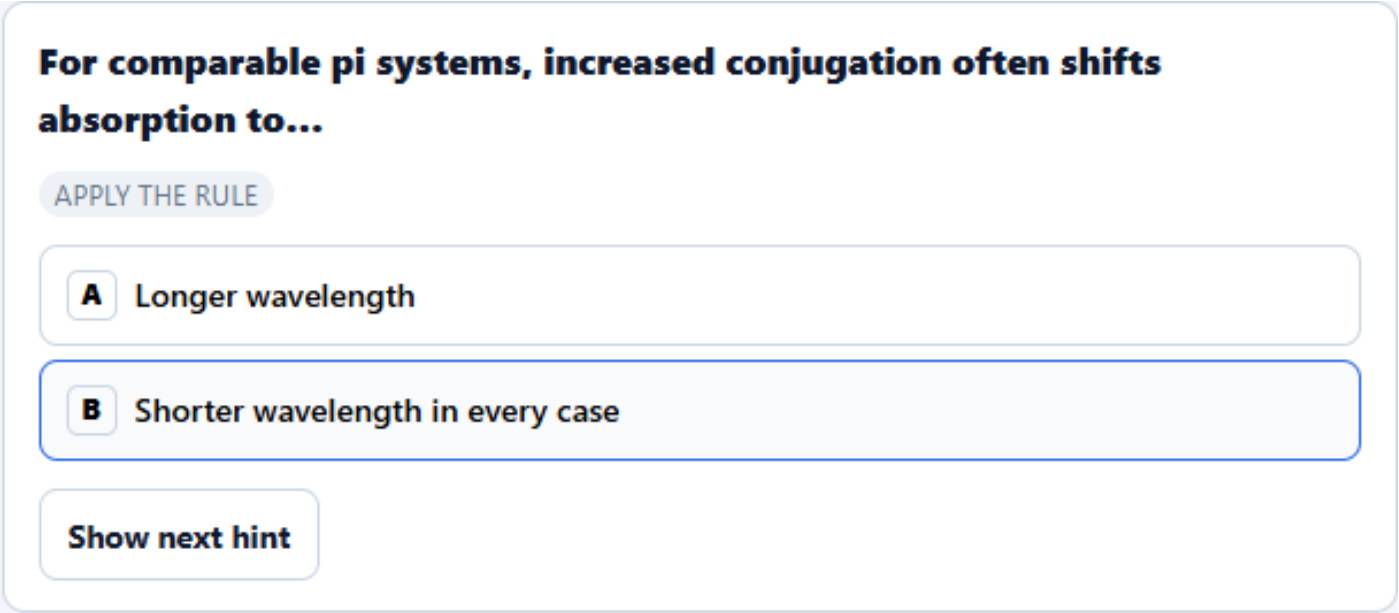
Option 0 (A): Longer wavelength

Option 1 (B): Shorter wavelength in every case

Screenshot: uv-energy-stage-2-molecule



Screenshot: uv-energy-stage-2-question



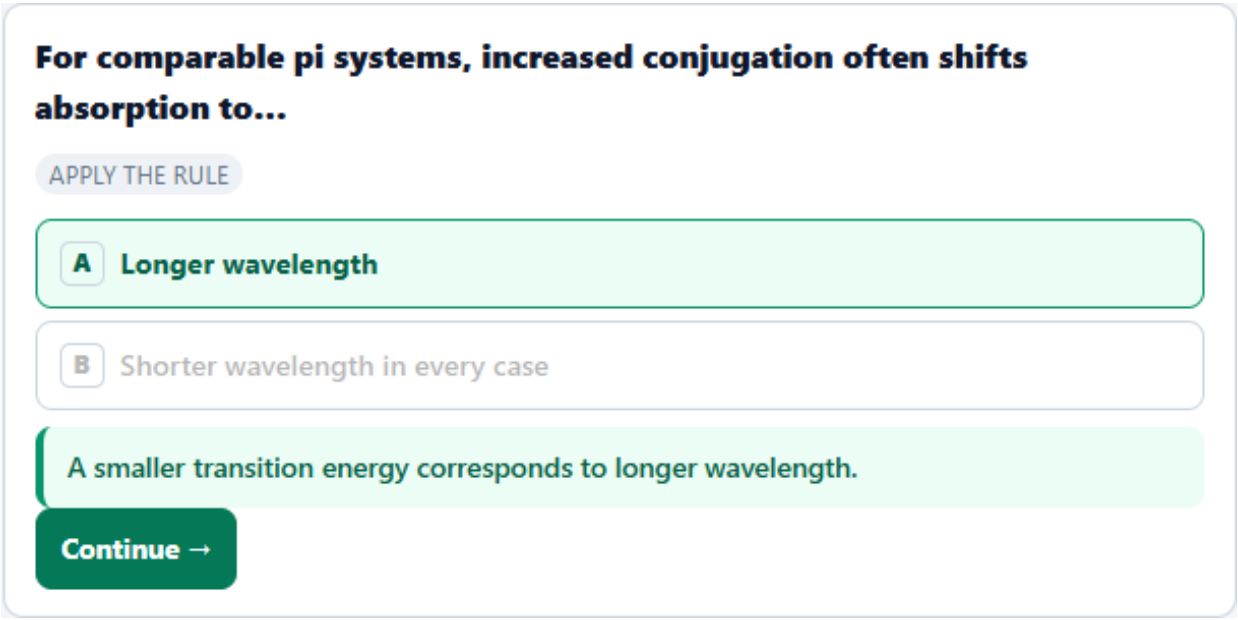
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use E=hc/λ.

Hint 2: Lower energy and longer wavelength are inversely related.

Explanation: A smaller transition energy corresponds to longer wavelength.

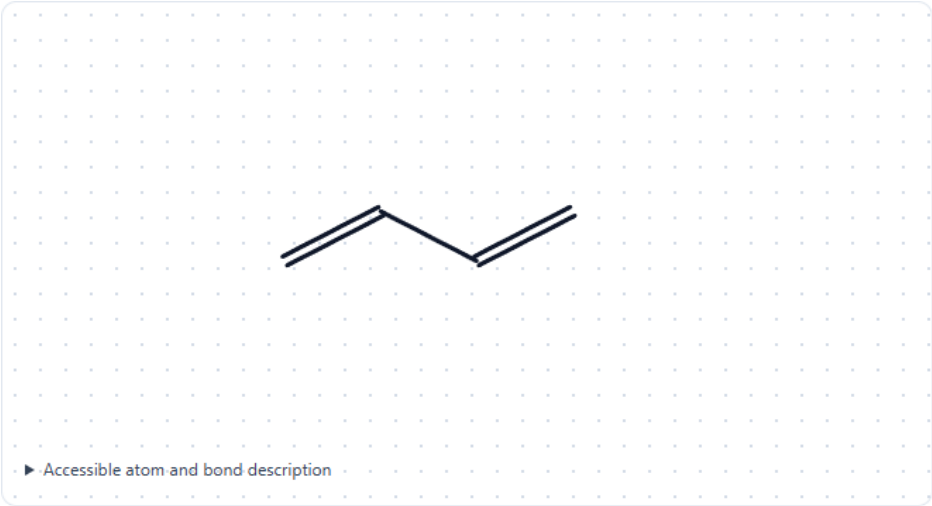
Screenshot: uv-energy-stage-2-feedback



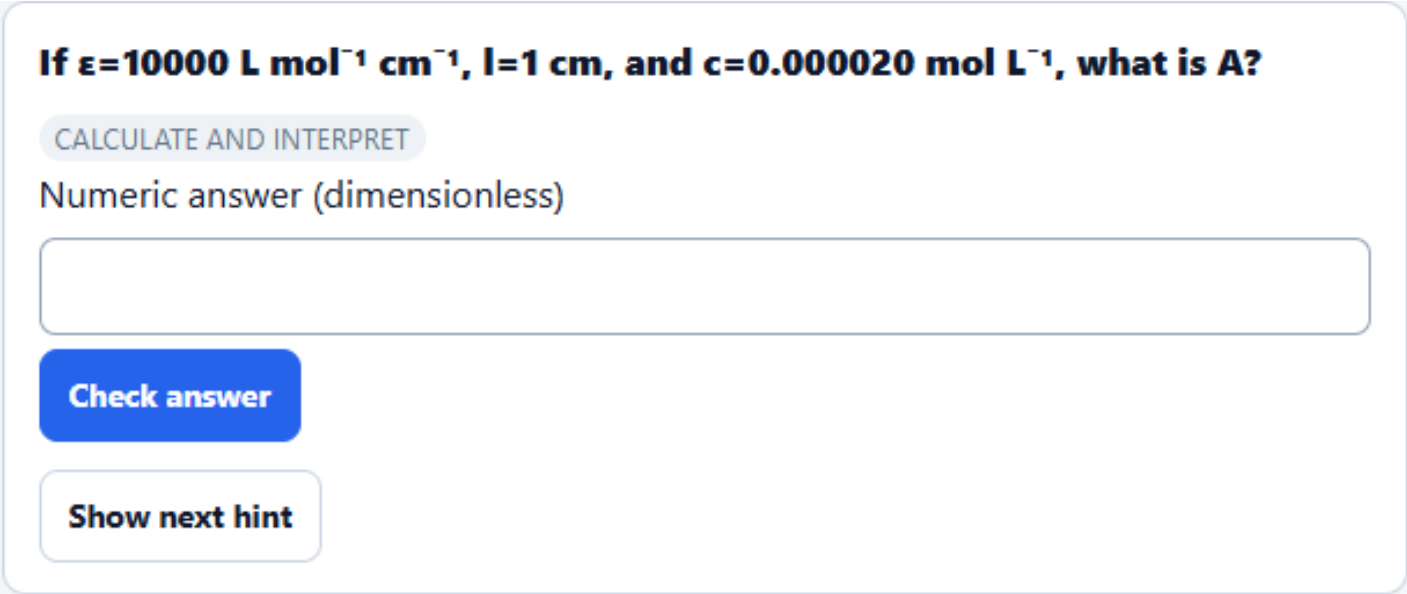
uv-energy:stage-3 | If ε=10000 L mol^{-1} cm^{-1}, l=1 cm, and c=0.000020 mol L^{-1}, what is A?

Answer type: number | unit: dimensionless

Screenshot: uv-energy-stage-3-molecule



Screenshot: uv-energy-stage-3-question



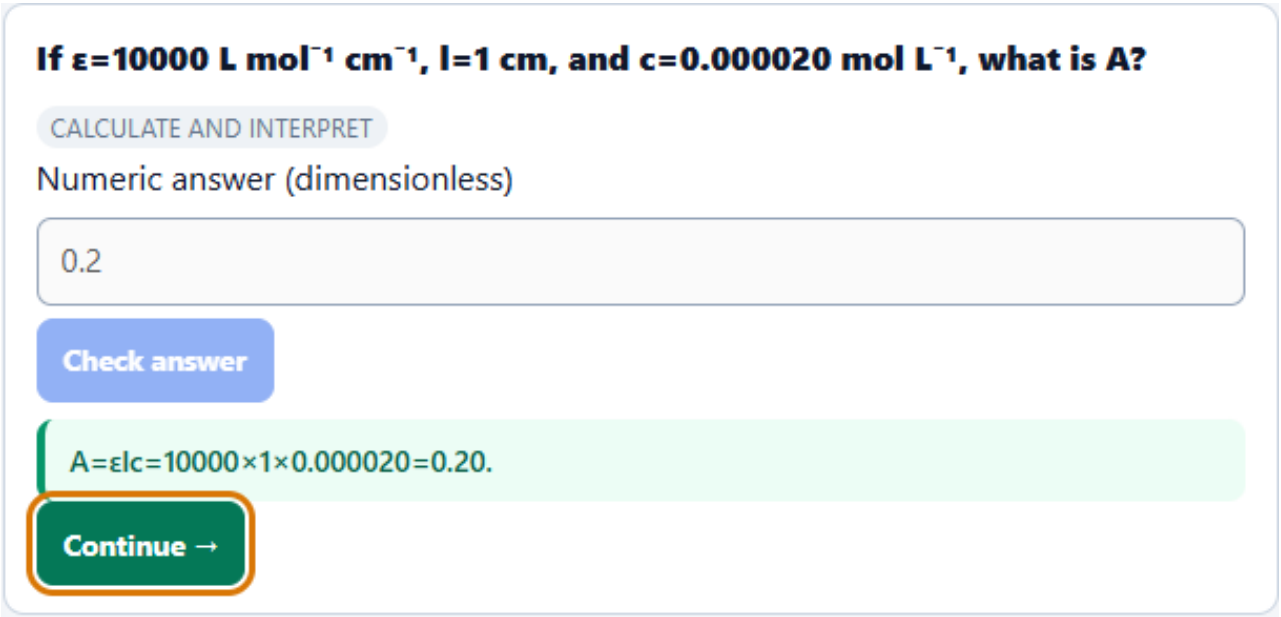
{"authorAnswer": 0.2, "tolerance": 0.001, "highlightsAfterCorrectAnswer": {}}

Hint 1: Multiply the three quantities.

Hint 2: The specified units cancel.

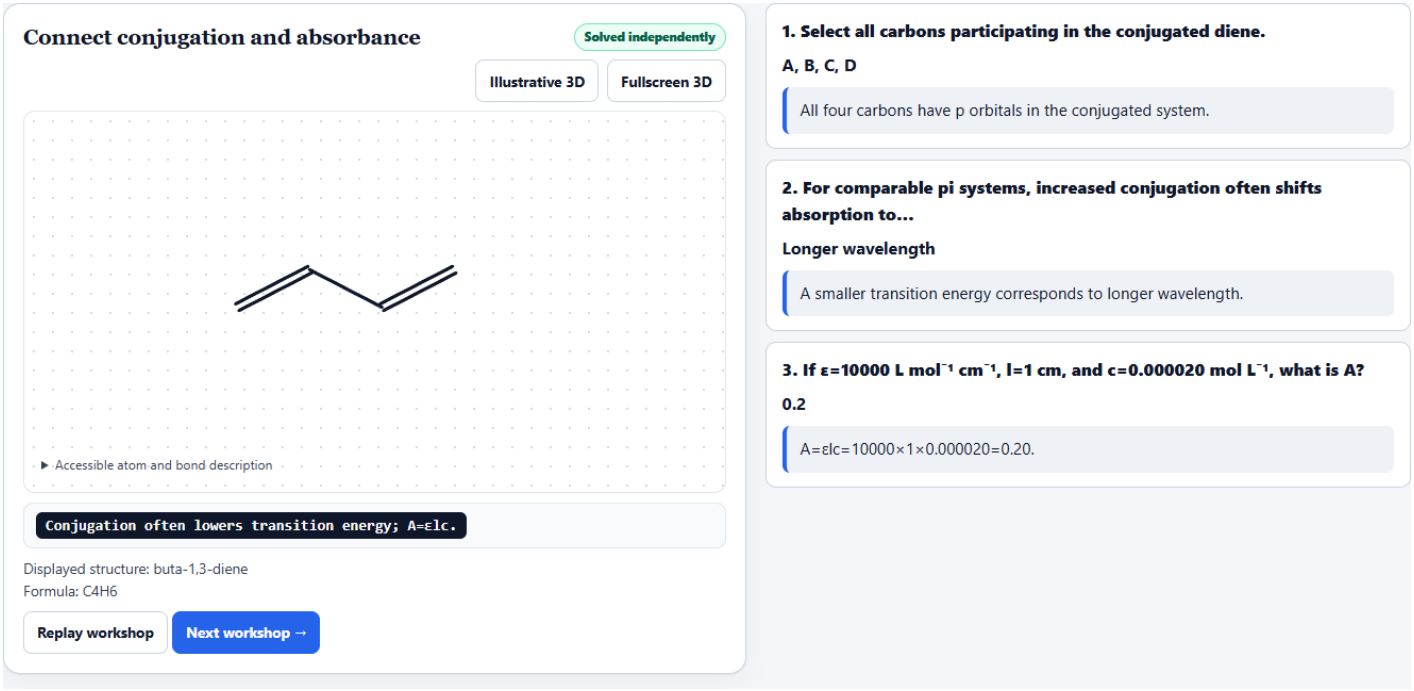
Explanation: $A = \varepsilon l c = 10000 \times 1 \times 0.000020 = 0.20$.

Screenshot: uv-energy-stage-3-feedback



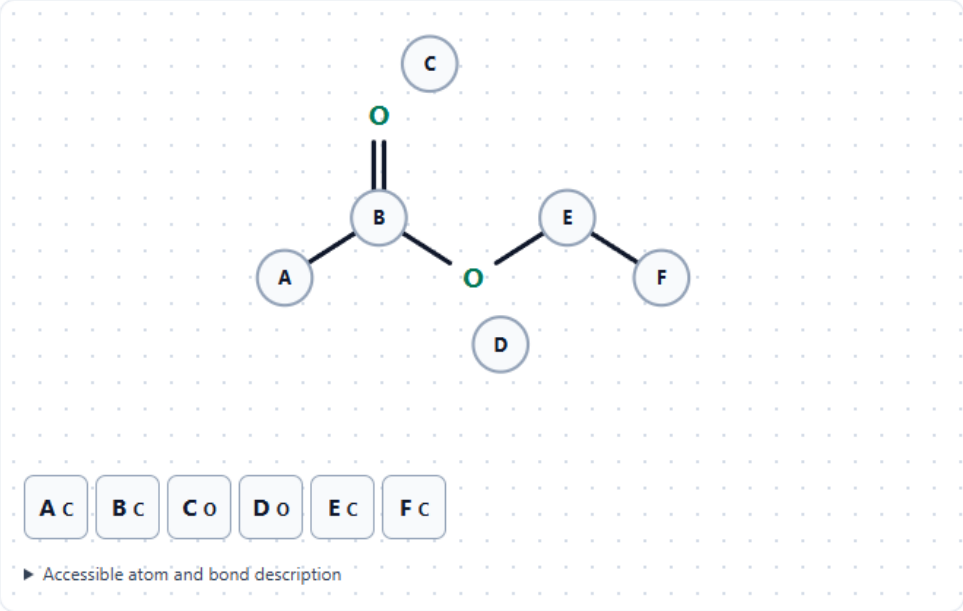
Completion summary: Conjugation often lowers transition energy; A=εlc.

Screenshot: uv-energy-complete

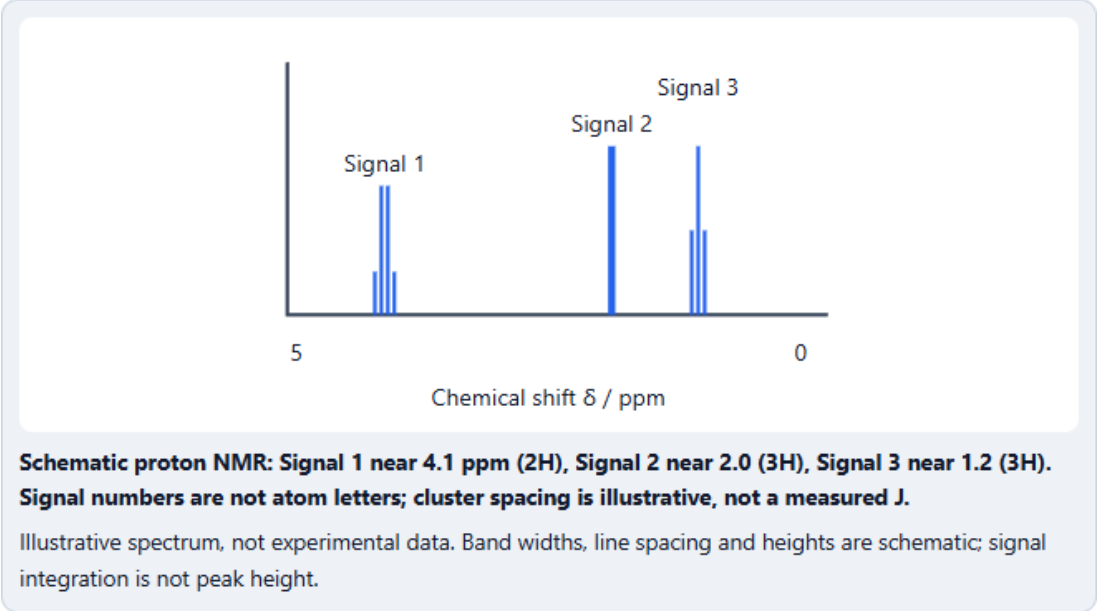


nmr-ethyl | Assign an ethyl-group pattern

Screenshot: nmr-ethyl-stage-1-molecule



Screenshot: nmr-ethyl-diagram

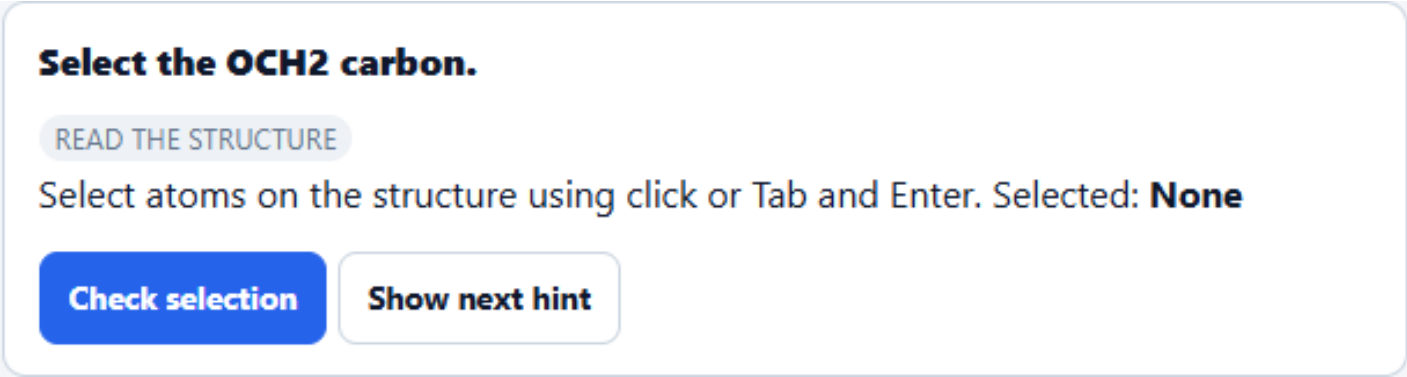


{"diagramSpecification": {"type": "spectrum", "min": 0, "max": 5, "unit": "Chemical shift δ / ppm", "caption": "Schematic proton NMR: Signal 1 near 4.1 ppm (2H), Signal 2 near 2.0 (3H), Signal 3 near 1.2 (3H). Signal numbers are not atom letters; cluster spacing is illustrative, not a measured J.", " peaks": [{"x": 4.1, "height": 0.65, "label": "Signal 1", "lines": 4, "spacing": 0.06}, {"x": 2, "height": 0.85, "label": "Signal 2", "lines": 1}, {"x": 1.2, "height": 0.85, "label": "Signal 3", "lines": 3, "spacing": 0.06}]}}

nmr-ethyl:stage-1 | Select the OCH2 carbon.

Answer type: atoms

Screenshot: nmr-ethyl-stage-1-question



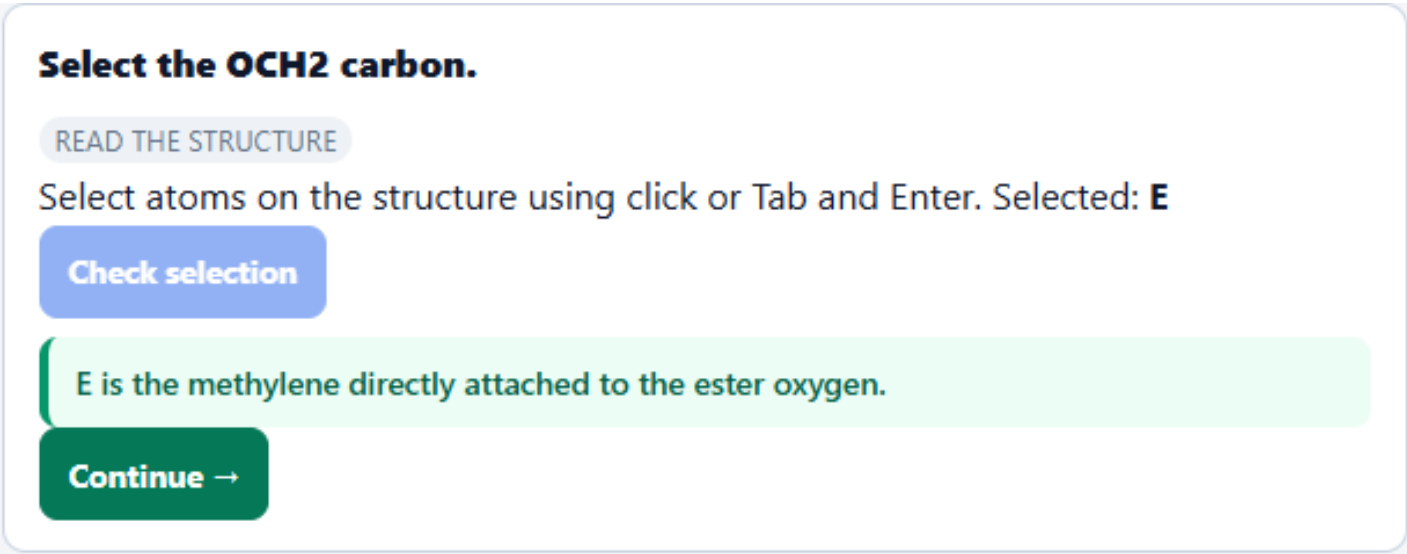
{"authorAnswer": [4], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [4], "sub": [5]}}

Hint 1: Find O–CH2–CH3.

Hint 2: Choose the carbon nearest O on the ethyl side.

Explanation: E is the methylene directly attached to the ester oxygen.

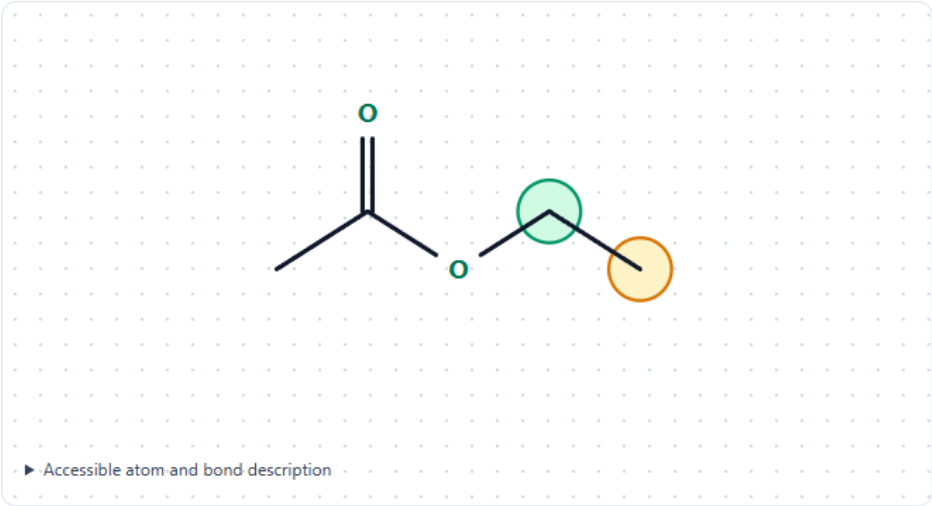
Screenshot: nmr-ethyl-stage-1-feedback



nmr-ethyl:stage-2 | How many lines does its signal have in the simple n+1 model?

Answer type: number | unit: lines

Screenshot: nmr-ethyl-stage-2-molecule



Screenshot: nmr-ethyl-stage-2-question

How many lines does its signal have in the simple n+1 model?

CALCULATE AND INTERPRET

Numeric answer (lines)

Check answer

Show next hint

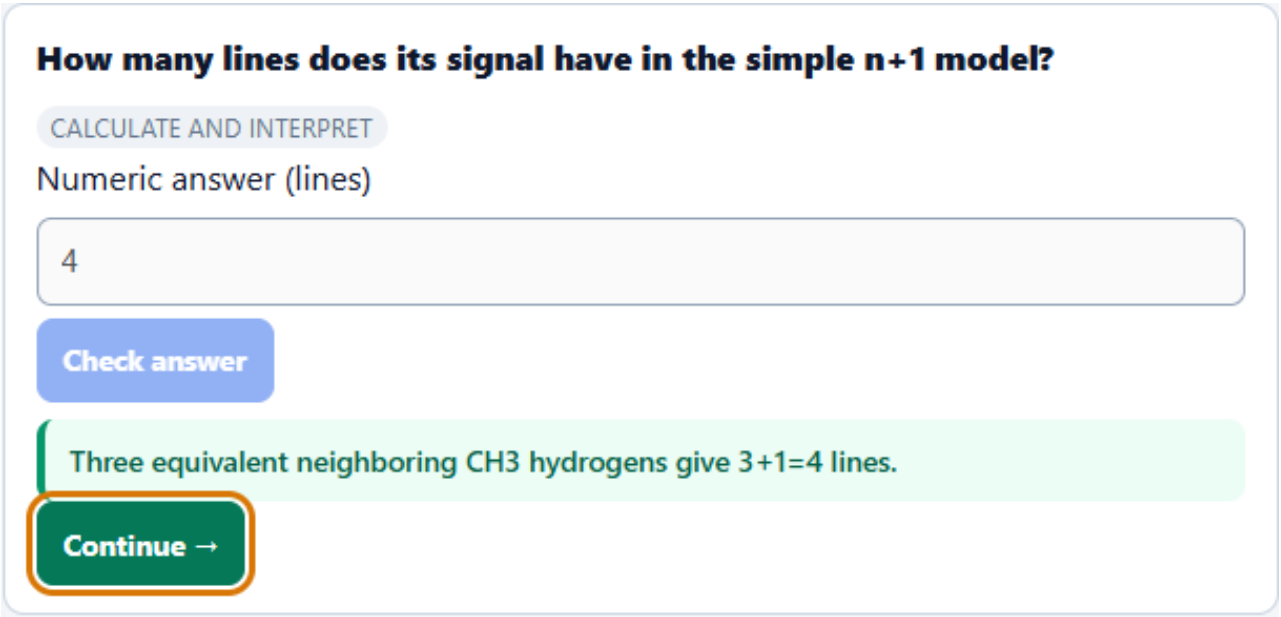
{"authorAnswer": 4, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Count hydrogens on the adjacent methyl.

Hint 2: Do not count the two H on the observed carbon itself.

Explanation: Three equivalent neighboring CH3 hydrogens give 3+1=4 lines.

Screenshot: nmr-ethyl-stage-2-feedback



nmr-ethyl:stage-3 | The terminal ethyl CH3 gives an approximate…

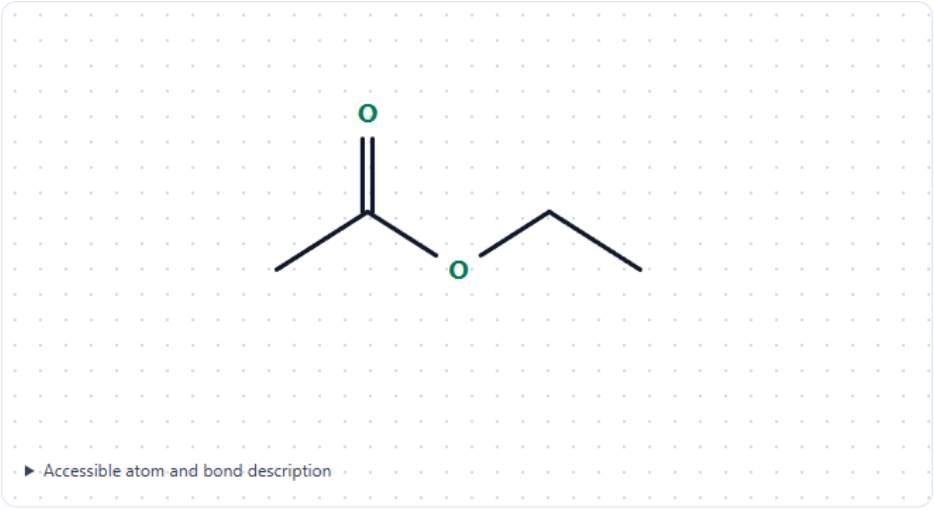
Answer type: choice

Option 0 (A): Triplet, integrating to 3H

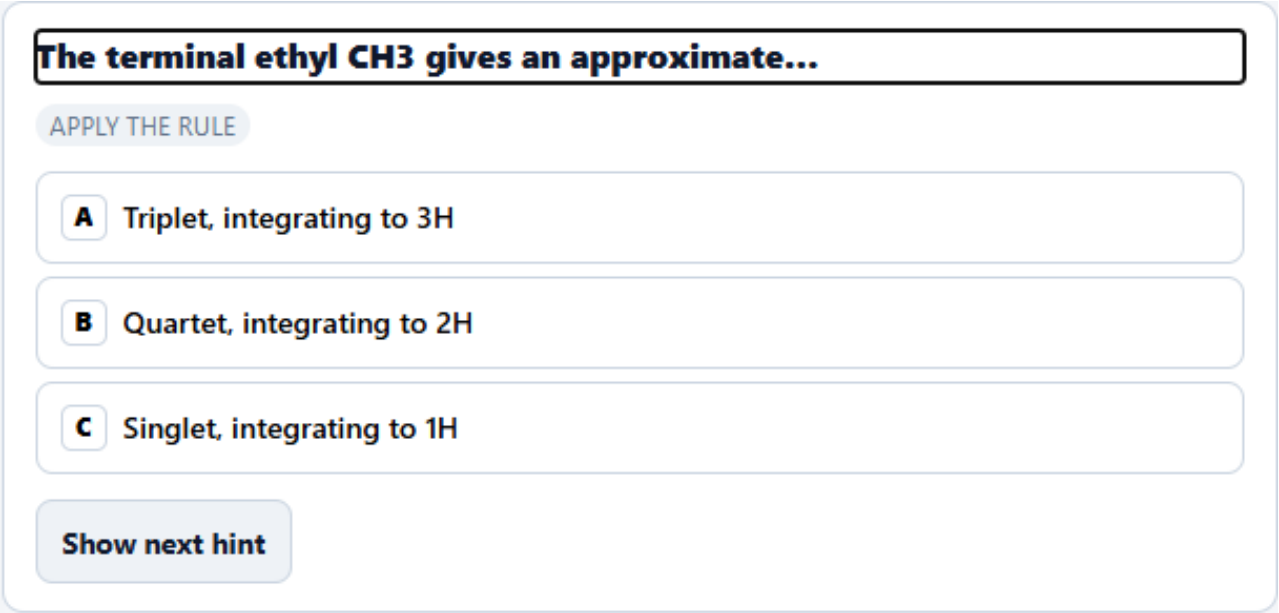
Option 1 (B): Quartet, integrating to 2H

Option 2 (C): Singlet, integrating to 1H

Screenshot: nmr-ethyl-stage-3-molecule



Screenshot: nmr-ethyl-stage-3-question



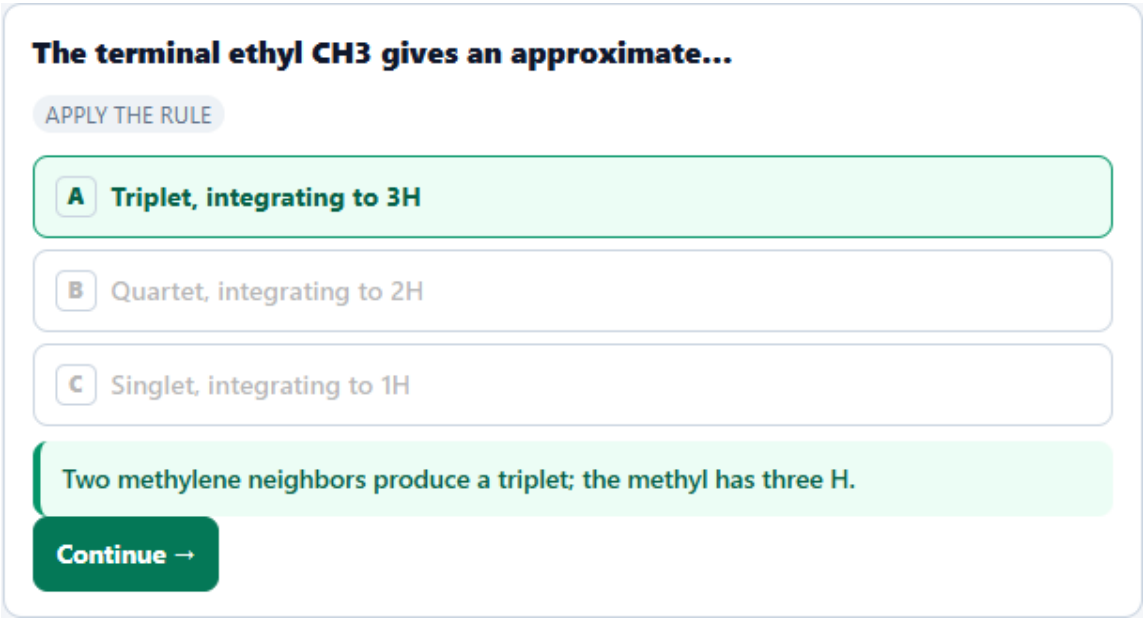
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Splitting and integration are separate measurements.

Hint 2: Use n=2 for the neighbors.

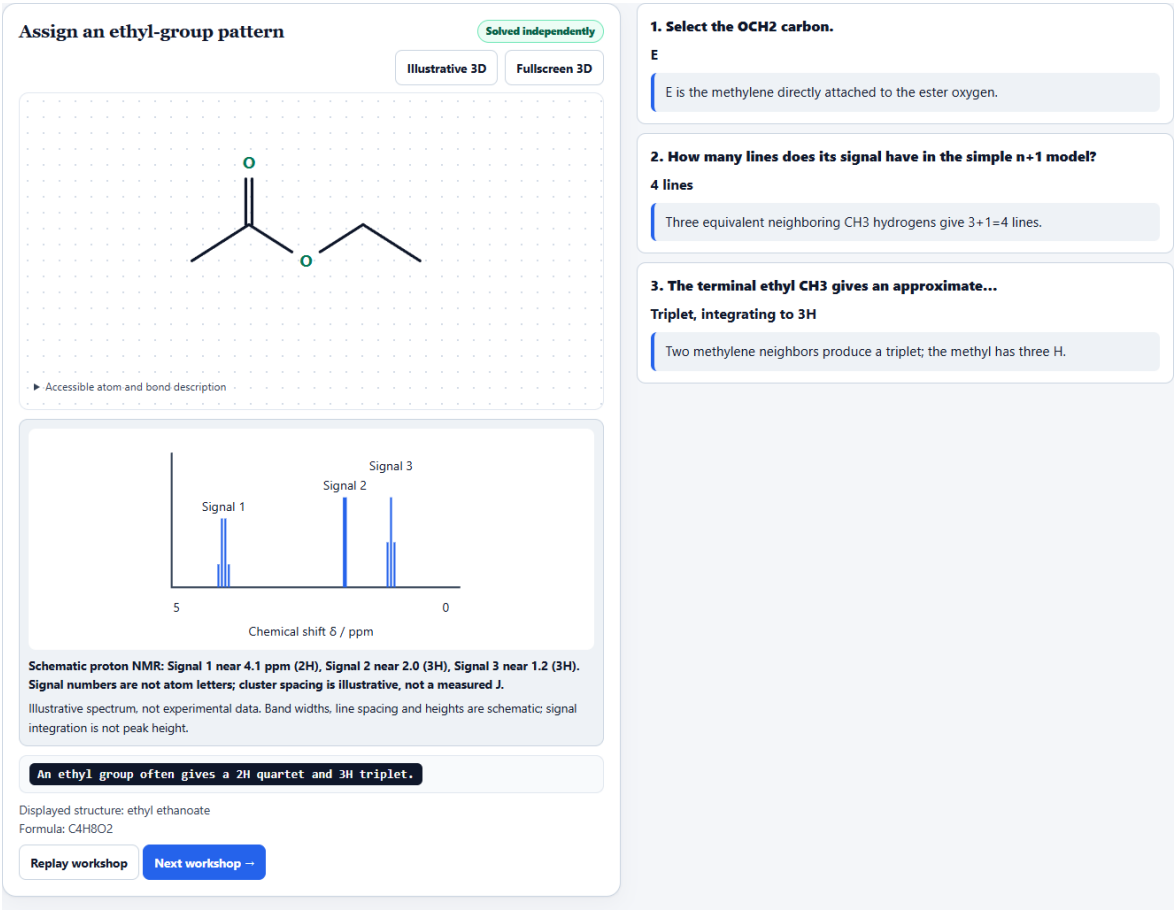
Explanation: Two methylene neighbors produce a triplet; the methyl has three H.

Screenshot: nmr-ethyl-stage-3-feedback



Completion summary: An ethyl group often gives a 2H quartet and 3H triplet.

Screenshot: nmr-ethyl-complete



combined-evidence | Test a structure against combined data

Supplied illustrative data: C3H6O; a ketone-like C=O band; one proton NMR singlet integrating to 6H near 2 ppm.

Screenshot: combined-evidence-stage-1-molecule

Candidate structure withheld. Use the supplied observations to infer it; the structure is shown after completion.

combined-evidence:stage-1 | For formula C3H6O, calculate DBE.

Answer type: number | unit: DBE

Screenshot: combined-evidence-stage-1-question

For formula C3H6O, calculate DBE.

CALCULATE AND INTERPRET

Numeric answer (DBE)

Check answer

Show next hint

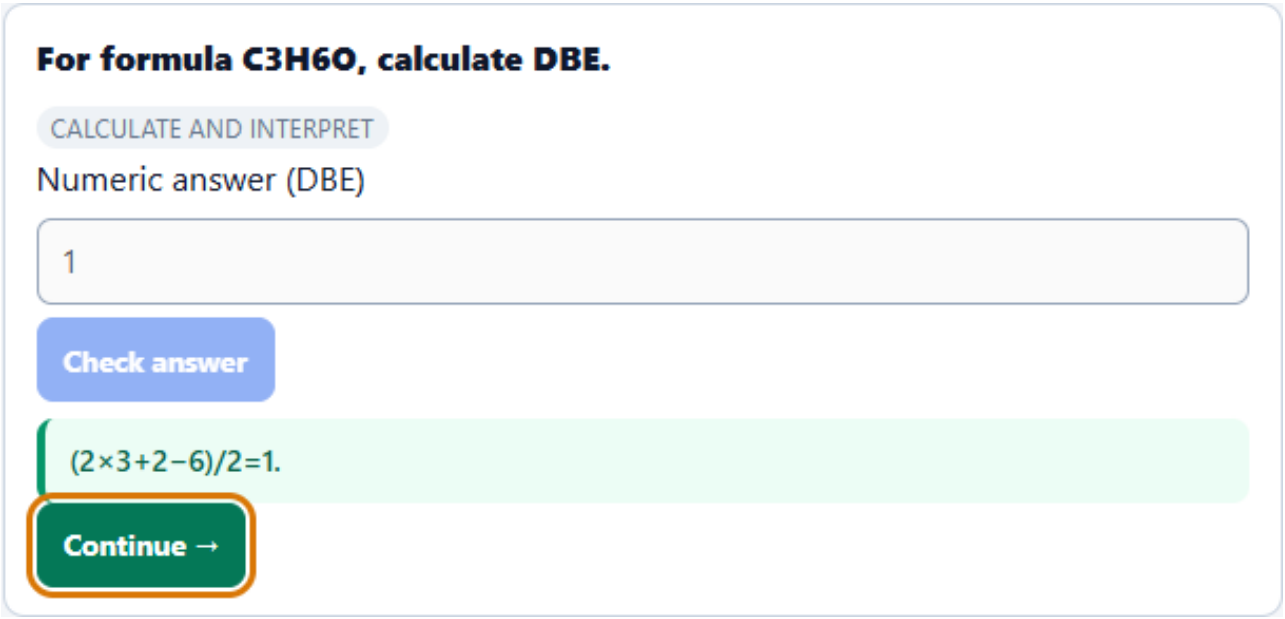
{"authorAnswer": 1, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Ignore O in this standard formula.

Hint 2: Compare with the saturated acyclic C3H8 formula.

Explanation: (2×3+2−6)/2=1.

Screenshot: combined-evidence-stage-1-feedback



combined-evidence:stage-2 | A single 6H singlet near 2 ppm suggests...

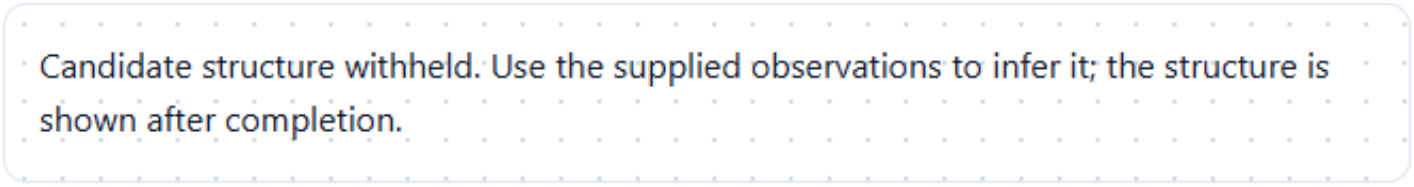
Answer type: choice

Option 0 (A): Two equivalent methyl groups with no adjacent H-bearing carbon

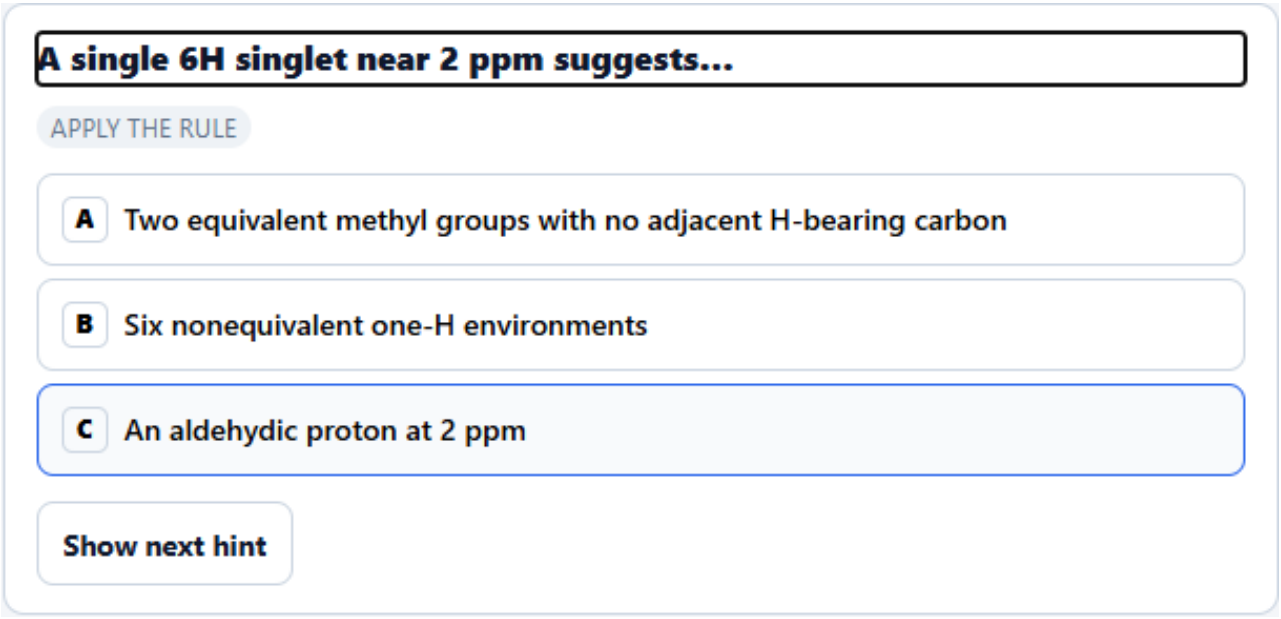
Option 1 (B): Six nonequivalent one-H environments

Option 2 (C): An aldehydic proton at 2 ppm

Screenshot: combined-evidence-stage-2-molecule



Screenshot: combined-evidence-stage-2-question



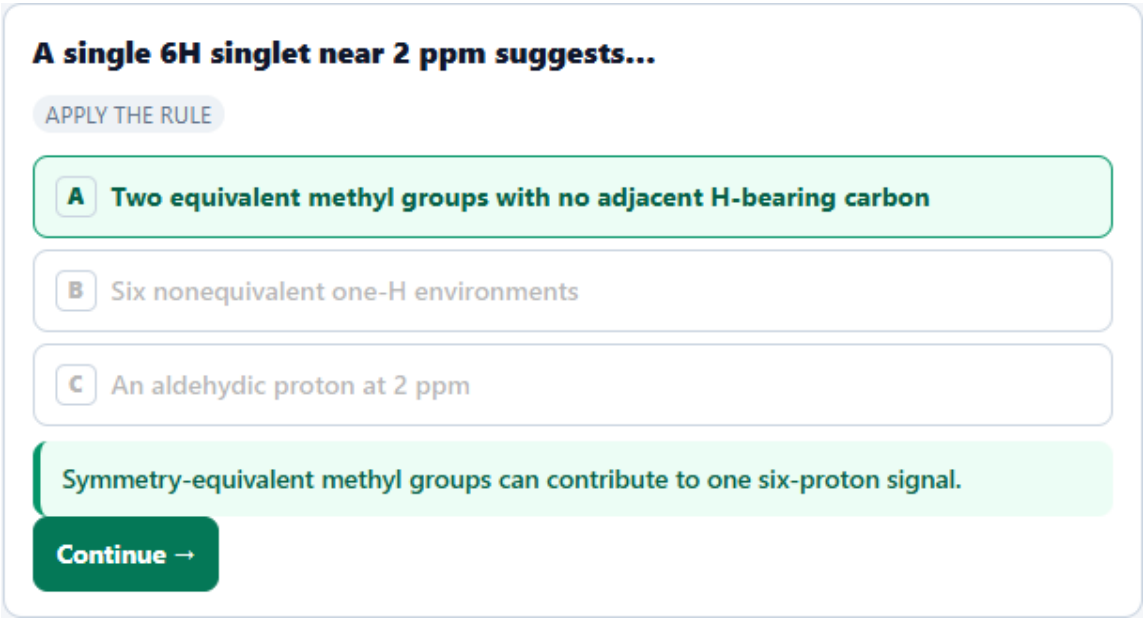
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Integration counts all protons in the environment.

Hint 2: An aldehyde H would usually be much farther downfield.

Explanation: Symmetry-equivalent methyl groups can contribute to one six-proton signal.

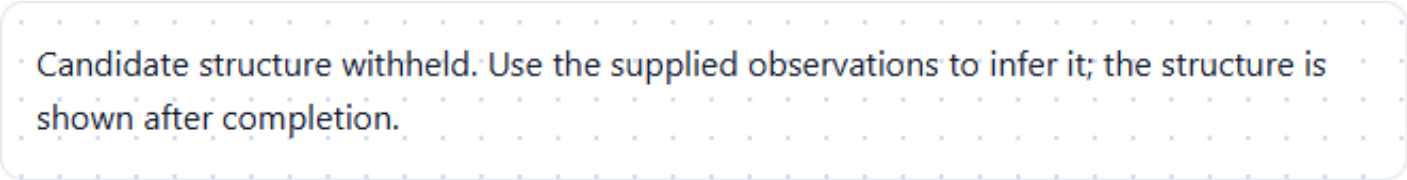
Screenshot: combined-evidence-stage-2-feedback



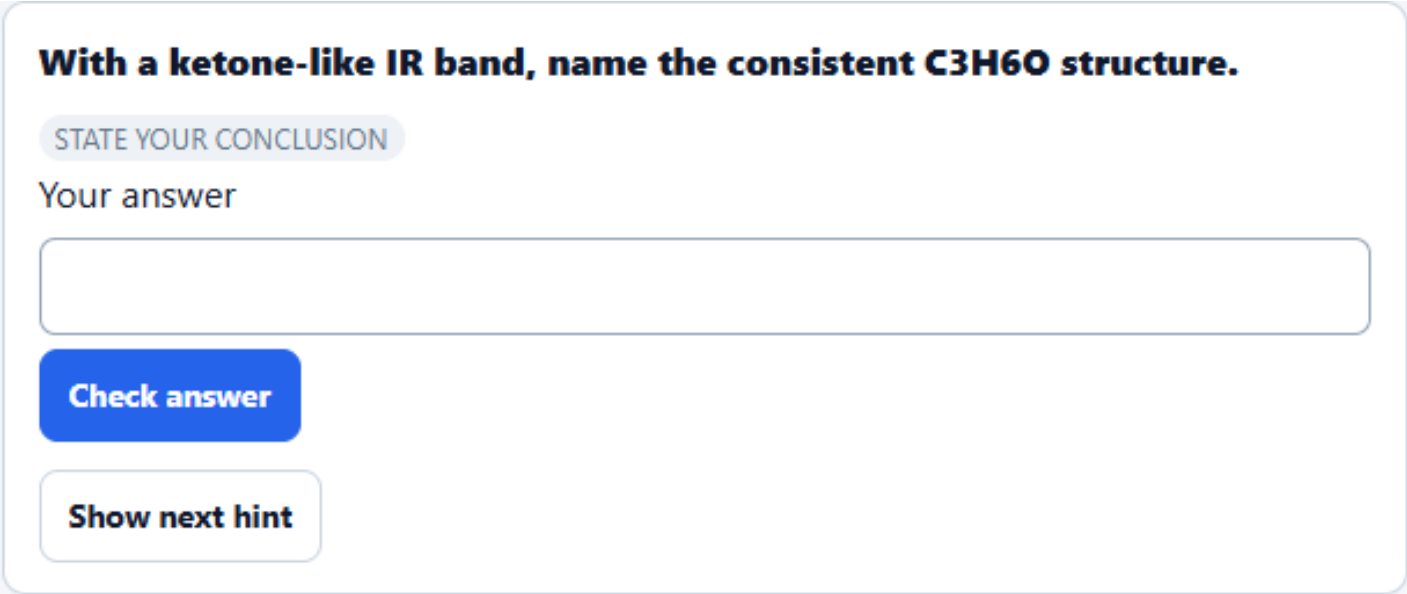
combined-evidence:stage-3 | With a ketone-like IR band, name the consistent C3H6O structure.

Answer type: text

Screenshot: combined-evidence-stage-3-molecule



Screenshot: combined-evidence-stage-3-question



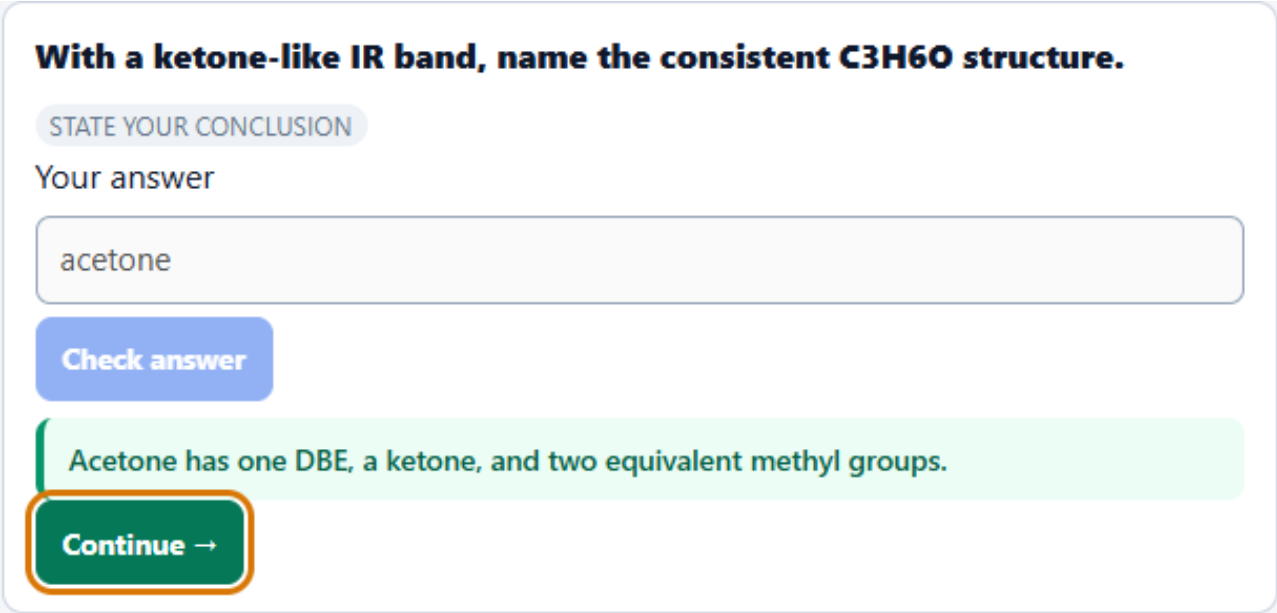
{"authorAnswer": ["acetone", "propanone", "propan-2-one", "2-propanone", "dimethyl ketone"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Place C=O between two methyl groups.

Hint 2: Check all three observations together.

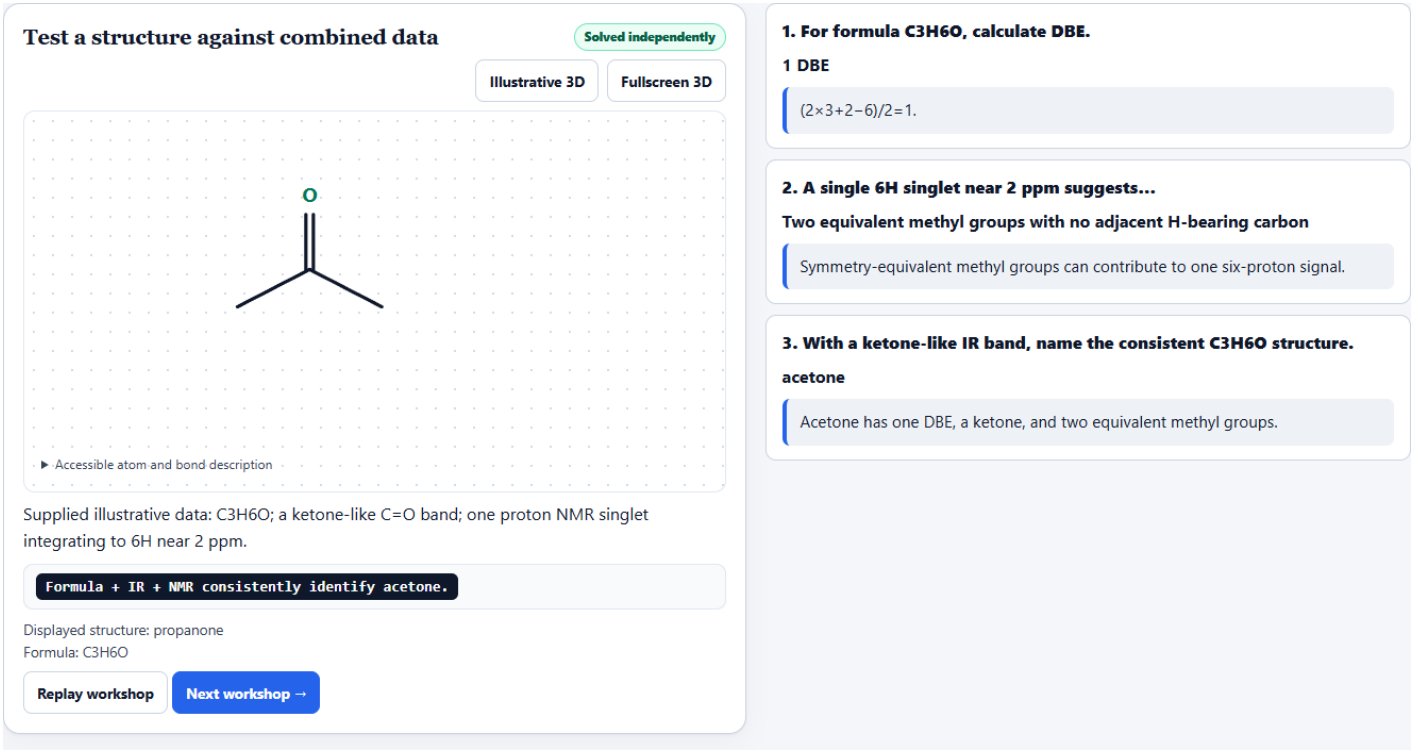
Explanation: Acetone has one DBE, a ketone, and two equivalent methyl groups.

Screenshot: combined-evidence-stage-3-feedback



Completion summary: Formula + IR + NMR consistently identify acetone.

Screenshot: combined-evidence-complete



Molecular repository: author claims to verify

acetone

Author claims: propanone | formula C3H6O | canonical composition C3H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

isopropanol

Author claims: propan-2-ol | formula C3H8O | canonical composition C3H8O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1]]

aceticacid

Author claims: ethanoic acid | formula C2H4O2 | canonical composition C2H4O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=108.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

butadiene

Author claims: buta-1,3-diene | formula C4H6 | canonical composition C4H6 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2]]

ethene

Author claims: ethene | formula C2H4 | canonical composition C2H4 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=131.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=189.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2]]

ethylacetate

Author claims: ethyl ethanoate | formula C4H8O2 | canonical composition C4H8O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=45.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=100.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=100.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=155.00, y=120.00; charge=0; explicit H override=none

4 (E): C; x=210.00, y=85.00; charge=0; explicit H override=none

5 (F): C; x=265.00, y=120.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1], [3, 4, 1], [4, 5, 1]]

propanal

Author claims: propanal | formula C3H6O | canonical composition C3H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=95.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=155.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=215.00, y=115.00; charge=0; explicit H override=none

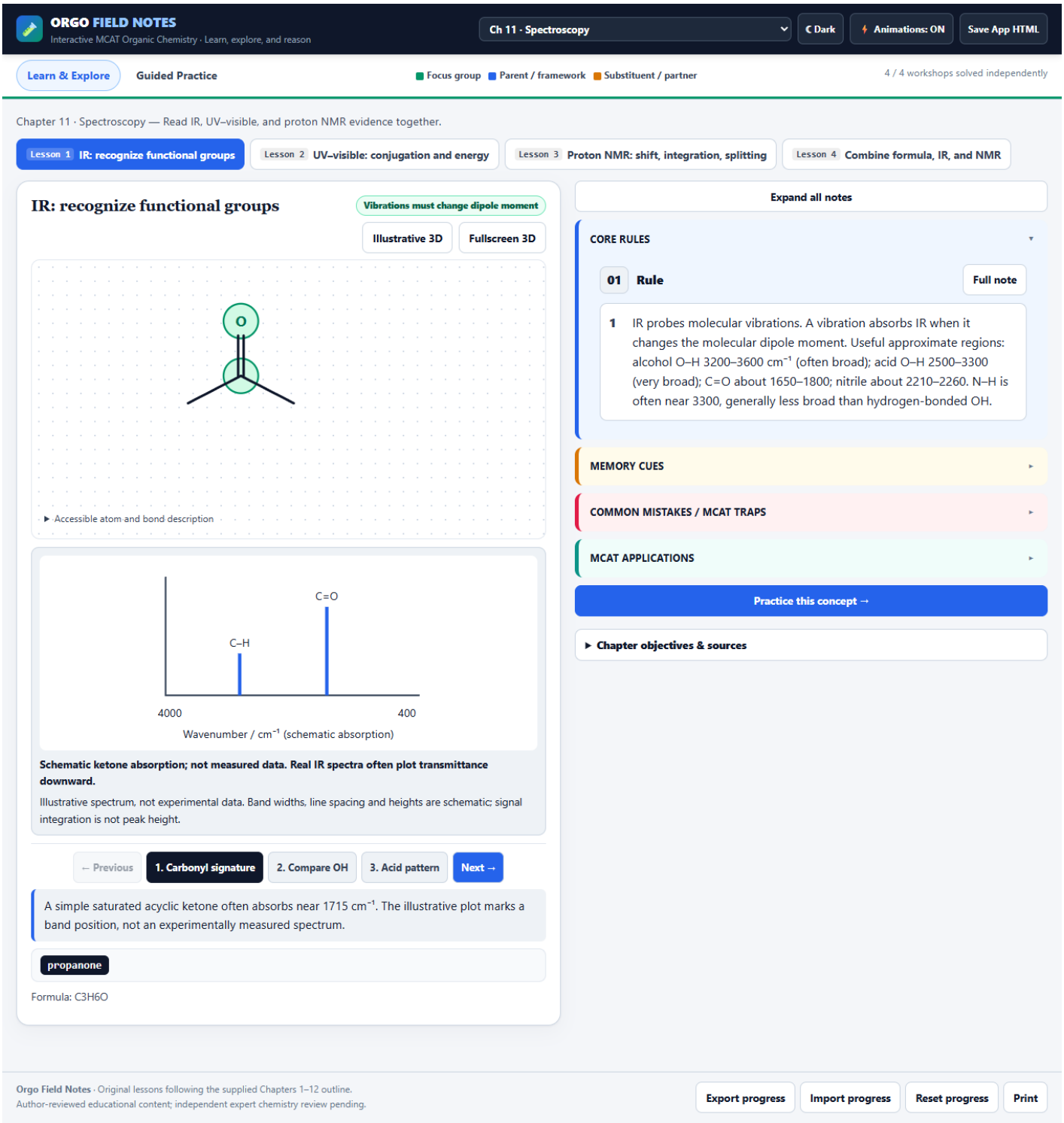
3 (D): O; x=267.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2]]

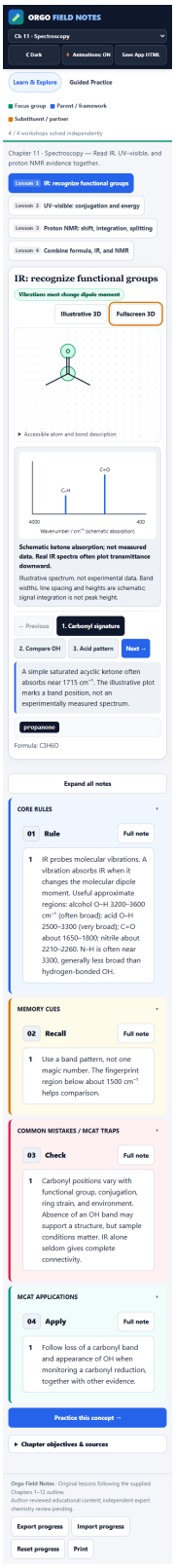
Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

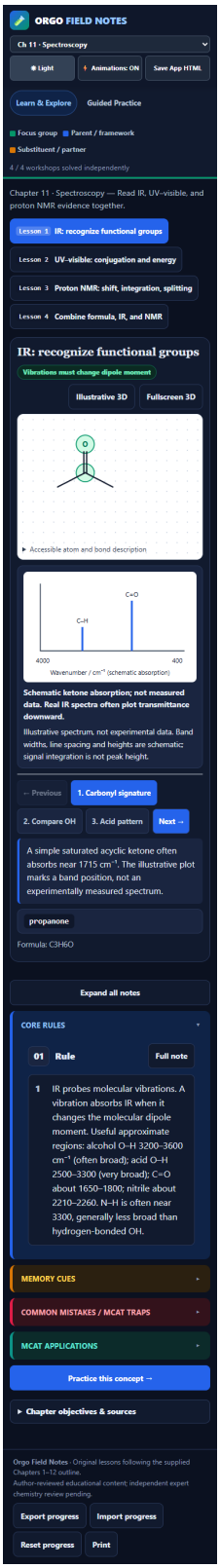
Screenshot: layout-desktop-light



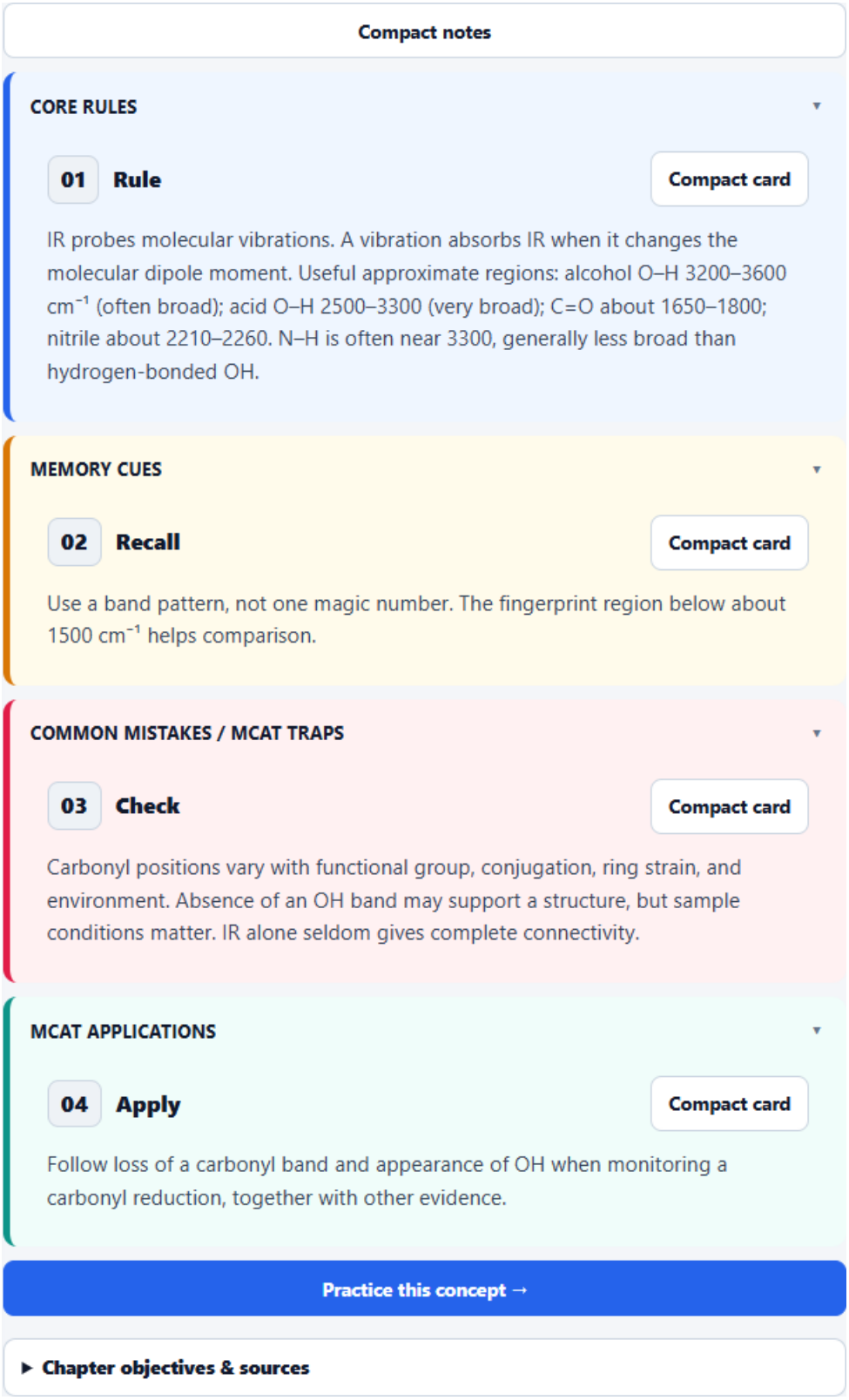
Screenshot: layout-mobile-light



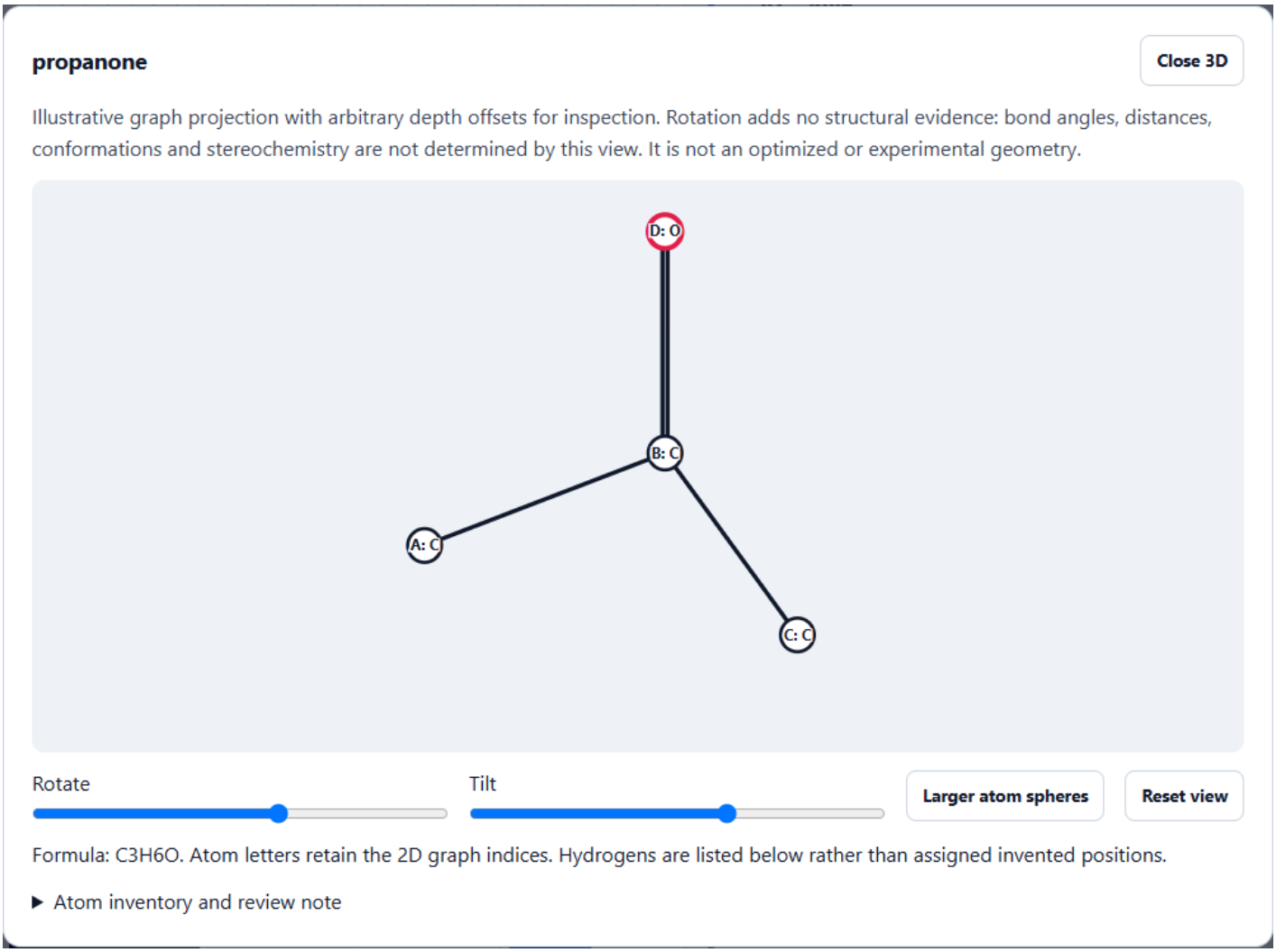
Screenshot: layout-mobile-dark



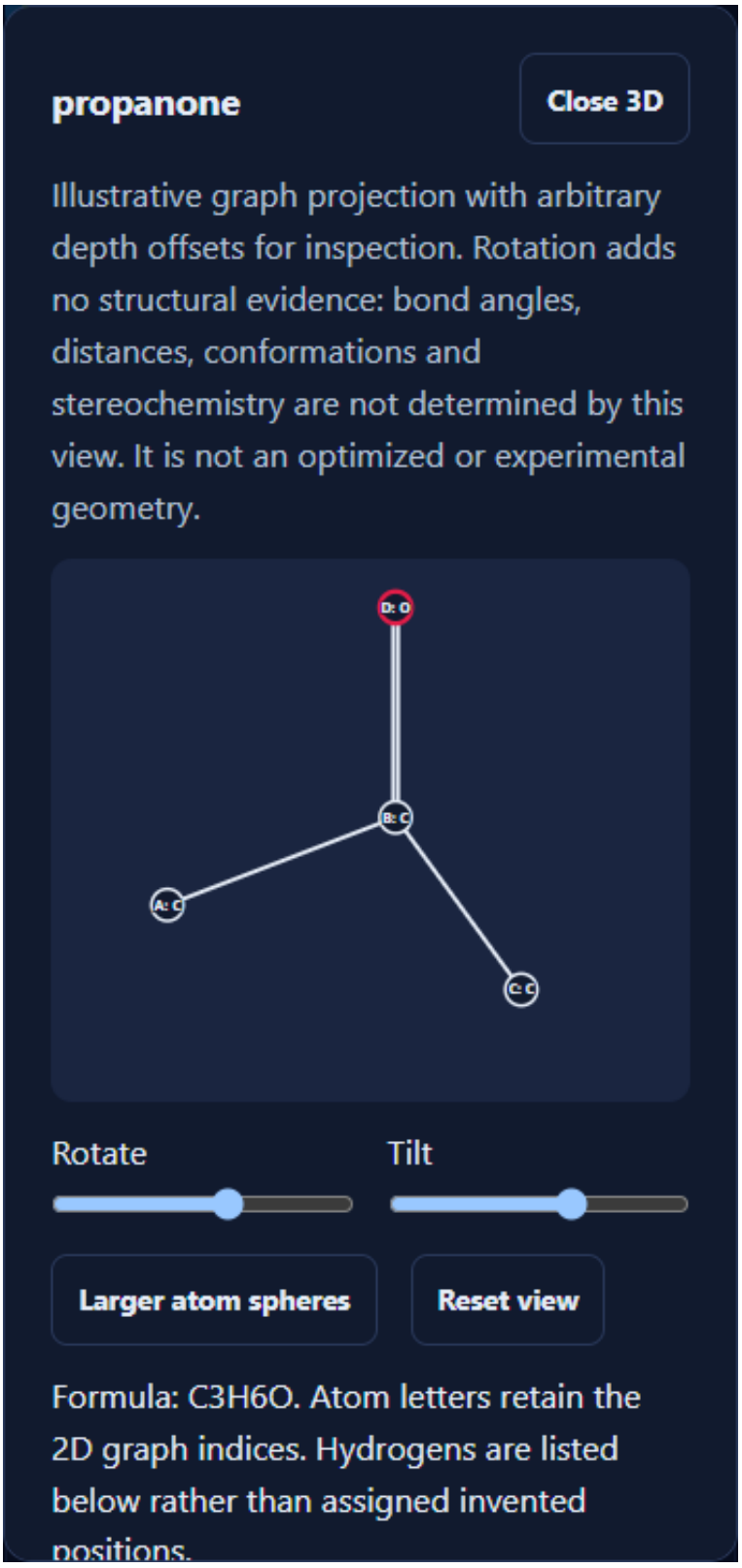
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

65 screenshots and 12 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.